

Readability Does Not Confer Control over Clinical Concepts

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Abstract

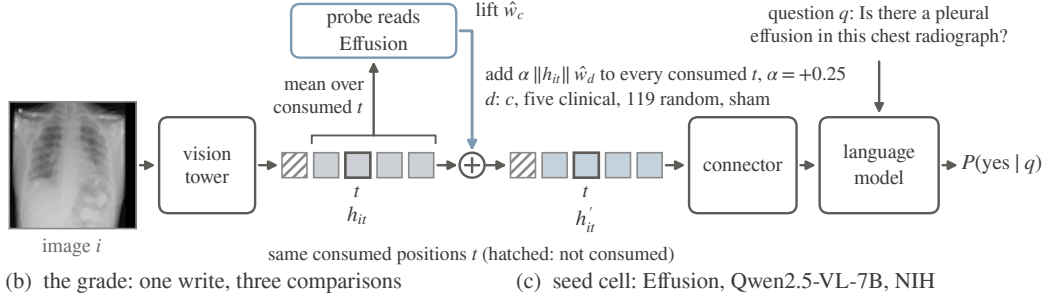
A linear probe reads a clinical finding from the visual representation that a vision-language model’s language model consumes, and activation steering suggests the converse: writing the probe direction back should move the model’s answer about the finding, making the direction a handle on the concept. Evaluations test whether models encode findings and answer questions about them, not whether such a handle exists. We test it with *ownership*: at a fixed dose on the same patients, does a concept’s own direction move its own answer more than the directions fitted for every competing finding? Across 25 checkpoints on NIH ChestX-ray14, CheXpert Plus and COCO, clinical concepts are readable in 59% of chest checkpoint–concept cells and answerable in 66%, but owned in 13%: most clinical writes do not move their own answer beyond random directions of the same dose, and the same directions move the other findings’ answers. Under the same procedure, 75% of natural-image object cells are owned, and the gap persists among cells the model answers well (30% against 87%). Non-clinical attributes of the radiographs share the deficit, and checkpoints that share a vision tower, and therefore write identical vectors, own different concepts: the deficit follows the chest-radiograph representation and the model that reads it. A direction fitted to the model’s own answers provides the handle that the label direction lacks. A probe’s readability does not confer control through its direction.

1 Introduction

Linear probes are the standard tool for testing what a vision-language model (VLM) represents. They read out a concept by fitting a classifier to activations at a chosen layer (Alain and Bengio, 2017; Belinkov, 2022). In medical imaging, probes often succeed at linearly decoding clinical findings such as pleural effusion or cardiomegaly from the visual representation of general-purpose and medical VLMs (Zhu et al., 2026; Pedersen et al., 2026). Activation steering promises the converse: writing the probe direction back into the representation to change the model’s behaviour (Turner et al., 2023; Panickssery et al., 2024). Together, these methods cast the probe direction as a handle on the concept: read it to detect, write it to control. We ask whether that handle exists.

A steering write that changes the answer does not settle the question. In Qwen2.5-VL-7B on NIH ChestX-ray14, a probe reads effusion from the visual tokens the language model consumes, and writing its direction back raises the model’s probability of answering yes to the Effusion question by 0.190, more than the 95th percentile of 119 random directions and a coordinate-permutation sham written at the same dose. By the usual steering criteria, the write works. Writing the Nodule direction into the same patients at the same dose raises the Effusion answer further, by 0.262 (Figure 1c), although Nodule is the competing direction least similar to Effusion (Appendix A.7). Random and sham directions carry no label, so they cannot expose a direction whose effect is shared with other

(a) read and write at the final visual block the language model consumes



(b) the grade: one write, three comparisons

(c) seed cell: Effusion, Qwen2.5-VL-7B, NIH

$W_{q,d}$: mean change in $P(\text{yes} | q)$ under the write of d

- own effect
 $W_{q,q} > 0$ ✓
- random and sham
 $W_{q,q} > \text{random p95 and } > \text{lshaml}$ ✓
- five clinical directions
 $W_{q,q} - W_{q,d} > 0$ for every $d \neq q$ ✗

owned: all three hold under simultaneous max- T bounds
seed cell: stronger competitor

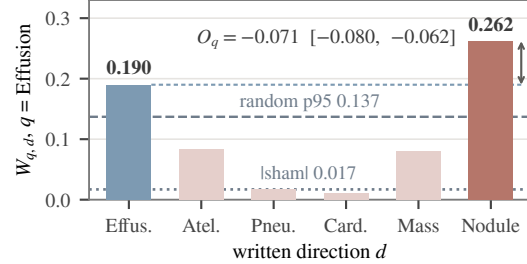


Figure 1: Reading and writing a concept at the consumed block. (a) The probe reads the concept from the consumed visual tokens, and its lifted normal is written back to them at dose $\alpha = +0.25$, a quarter of each token’s norm. (b) The write matrix $W_{q,d}$, the mean change in $P(\text{yes})$ of question q under a write along direction d , and the three comparisons that decide ownership. (c) The Effusion case (Qwen2.5-VL-7B, NIH ChestX-ray14): a stronger competitor.

findings (Pahde et al., 2025). Exposing it requires writing every competing clinical direction at the same locus and dose in the same patients and asking whether the concept’s own direction moves the concept’s own answer more than any other. We call the outcome *ownership*. When a concept is readable but not owned, its probe direction reads the concept without writing it: reading is not writing.

We turn this comparison into an evaluation (Section 3, Figure 1). At the *consumed block*, the final visual block whose output the language model reads, each concept receives three grades. It is *readable* if its probe beats random-label probes of equal capacity, *answerable* if the model’s yes/no answer without any write beats chance, and *owned* if a fixed-dose write along its direction clears the random and sham references and moves its own answer more than every competing clinical direction under simultaneous bounds. We apply the evaluation to 25 checkpoints from twelve families (3B–72B; general-purpose and medical) on two chest-radiograph datasets with report-derived labels, NIH ChestX-ray14 and CheXpert Plus, and on COCO objects as a natural-image control (Section 4).

On chest radiographs most readable and answerable concepts are not owned, while most COCO objects are, and the gap persists among concepts the model answers well (Section 5). We call this gap the *attribution deficit*. Since the same write succeeds on natural images, the remaining results locate the deficit. Checkpoints that share a vision tower write identical vectors into identical representations yet own different concepts, so the model that reads the representation decides ownership. Non-clinical attributes of the same radiographs (projection, sex, and age) are owned no more often than findings, so the deficit does not depend on the concept being clinical. The reader responds to a different direction: a direction fitted to the model’s own answers is owned in 19 of 36 chest checkpoint–concept cells where the label direction is owned in 9, and on chest radiographs it is nearly orthogonal to the probe direction.

Contributions. (i) We introduce ownership, which separates reading a concept from writing it: a probe direction is scored against directions fitted the same way for the competing concepts, written at the same locus and dose in the same patients (Section 3). (ii) We identify the attribution deficit:

across 450 checkpoint–concept cells, clinical concepts on chest radiographs are read and answered far more often than they are owned, and the gap to natural-image objects persists at matched answer ability (Sections 5.1–5.3). (iii) We trace the deficit to how the model reads the chest-radiograph representation rather than to the concept’s clinical status, and show that a direction fitted to the model’s own answers provides the handle that the label direction lacks (Sections 5.4–5.6). We release the evaluation, scorecards, and write matrices; a new model needs one adapter naming its consumed block.

2 Background and Related Work

Probing and its controls. Linear probes track information across network depth (Alain and Bengio, 2017), and a probe score reflects both the representation and the learner fitted to it, which motivates matched-capacity probes and control tasks (Hewitt and Liang, 2019). The probing literature distinguishes information a readout can recover from information the model uses (Belinkov, 2022; Ravichander et al., 2021); causal erasure tests use by deleting a concept (Ravfogel et al., 2020; Elazar et al., 2021), in closed form for every linear reader at once (Belrose et al., 2023). Our readability grade adopts the control-task idea, and the write test asks a question complementary to erasure: whether the concept’s own direction is the one that moves its answer. In medical multimodal models, depth-wise probing maps classification information across towers, connectors, and language layers (Zhu et al., 2026), and layer-wise probes of MedCLIP expose acquisition and device shortcuts (Pedersen et al., 2026); we intervene on the decoded clinical direction at the locus where it is read.

Concept directions and steering. A concept activation vector is the normal of a classifier separating concept examples from a negative set, and it grades a class by the directional derivative of the logit along that normal (Kim et al., 2018); we write the direction forward instead. Such a normal shifts with the layer it is fitted at, entangles with correlated concepts, depends on its negative set, and carries where in the image the concept was read (Nicolson et al., 2025). Representation engineering and activation steering write a concept back by changing intermediate representations at inference time (Zou et al., 2023; Turner et al., 2023; Panickssery et al., 2024), and a probe normal is an appealing intervention because it supplies a label and a signed vector from the same representation. A separability-trained vector can, however, absorb distractor signals and diverge from its concept (Pahde et al., 2025): a classifier’s weights are a filter rather than the pattern they detect (Haufe et al., 2014), mass-mean directions steer truth judgements better than logistic normals (Marks and Tegmark, 2024), and the features that identify a concept are largely not those that control the output (Arad et al., 2025). Benchmarks accordingly score concept detection and steering as separate axes (Wu et al., 2025), and alignment search certifies a subspace as causal when it implements an interpretable variable (Geiger et al., 2024). These works judge a direction mainly against its own concept. Ownership fixes the estimator and varies the label instead: it asks whether a direction fitted the same way for another concept moves the same answer more. Section 5.7 runs alternative constructions, a second locus, and token-concentrated writes through this test.

Writing into a vision-language model. Image features are linearly decodable in a vision-language model’s hidden layers, and targeted edits to them change downstream answers (Rajaram et al., 2025). Steering one sparse-autoencoder feature of the vision tower and asking the language model what it sees recovers what that feature encodes (Ferrando et al., 2026); we start from the meaning, fit a direction to a clinical label, and ask whether the write keeps that label’s selectivity against the other findings’ directions. Causal probing of multimodal models intervenes on the language model’s residual stream with difference-in-means vectors and scores each concept against its own effect (Deng et al., 2026); we write the representation the language model consumes and score each concept against its competitors.

3 Reading, Answering, and Writing at the Consumed Block

We ask three questions about a concept c in a checkpoint M on a dataset with binary labels: whether c is readable from the visual representation the language model consumes, whether M answers the question about c , and whether writing c 's own direction moves c 's answer more than any alternative. A registered protocol fixes an estimator and a decision rule for each question (Appendix A.1).

3.1 Consumed block and matched read–write positions

Every VLM in the grid passes an image through a vision tower and hands a specific tensor to a connector or merger that produces the language model's inputs. The *consumed block* is the last visual block whose output that consumer reads, restricted to the token positions the consumer actually uses (CLS or padding rows are excluded where the consumer drops them); the *connector locus* is the consumer's output. Appendix F.1 identifies the block for every family. For image i and consumed token t , let $\mathbf{h}_{it} \in \mathbb{R}^D$ be the output of the consumed block, where D is the tower's hidden width; mean pooling over the consumed tokens gives the read vector $\mathbf{h}_i$. A write adds the unit direction $\hat{\mathbf{w}}_c$ of concept c (Section 3.2) to the same tokens,

$$\mathbf{h}'_{it} = \mathbf{h}_{it} + \alpha \|\mathbf{h}_{it}\|_2 \hat{\mathbf{w}}_c, \quad (1)$$

so that the dose α is a fraction of each token's own norm, comparable across families whose activation scales differ by orders of magnitude, and its sign selects the concept's positive or negative pole. Reading and writing therefore use the same representation and the same positions. The connector and the language model, everything downstream of the consumed block, form the *reader* of that representation.

3.2 Directions, controls, and the reference family

Each concept is read by a linear probe on a shared 512-dimensional view of the representation. A fixed Gaussian matrix $R \in \mathbb{R}^{D \times 512}$ maps the read vector to features $\mathbf{x}_i = R^\top \mathbf{h}_i$, which are standardised with training-row means and scales s ; an ℓ_2 -regularised logistic regression with coefficient vector β_c then scores concept c (inverse regularisation strength 1, 2,000 iterations, three fit seeds). The probe's logit is thus an affine function of $\mathbf{h}_i$ whose gradient is $R \text{diag}(s)^{-1} \beta_c$, and the write direction is that gradient at unit length,

$$\hat{\mathbf{w}}_c = \frac{\mathbf{g}_c}{\|\mathbf{g}_c\|_2}, \quad \mathbf{g}_c = R \text{diag}(s)^{-1} \beta_c. \quad (2)$$

It is the direction in the consumed representation along which the probe's evidence for c grows fastest (Appendix F.3 gives the numerical details). We call it the concept's *label direction*, since it is fitted to the concept's labels.

Two families of controls bracket the label direction. For reading, 20 control probes are fitted to random labels drawn within recurring image types (view, sex, and age decade on chest sets; aspect and area buckets on COCO), so that they have the capacity and the nuisance structure of the concept probe. With y the concept labels and r the probe's scores on the calibration rows, and y_k^{ctrl} and r_k^{ctrl} the random labels and scores of control probe k , the probe's selectivity is

$$S = \text{AUROC}(y, r) - \frac{1}{20} \sum_{k=1}^{20} \text{AUROC}(y_k^{\text{ctrl}}, r_k^{\text{ctrl}}), \quad (3)$$

which is positive when the probe reads its concept better than a probe of equal capacity reads an arbitrary label with the same nuisance structure. For writing, the *reference family* contains 119 random unit directions, a coordinate-permutation sham of $\hat{\mathbf{w}}_c$ (same coordinates, scrambled arrangement), and the five label directions of the other clinical concepts, all written at the same dose to the same images.

3.3 Ownership

Each dataset has six concepts, and we index both the yes/no question about a concept and the direction fitted for it by that concept. For question q and direction d , $W_{q,d}$ is the mean over test images of the change in $P(\text{yes})$ for question q when $\hat{\mathbf{w}}_d$ is written, each image scored with and without the write. Here $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$, where σ is the logistic function and ℓ_{yes} and ℓ_{no} are the log-probabilities of the two answers at the first answer position, each the maximum over case and leading-space variants that tokenise to one token. Row q of the 6×6 write matrix W records how far each direction moves question q , and its diagonal entry $W_{q,q}$ is the concept’s own write (Figure 5). Ownership is the diagonal margin

$$O_q = W_{q,q} - \max_{d \neq q} W_{q,d}, \quad (4)$$

the advantage of the concept’s own direction over its strongest clinical competitor on the concept’s own question. $W_{q,q}$ measures how far the concept’s own direction moves its answer; O_q measures whether it moves that answer further than every competitor built the same way. A large $W_{q,q}$ shows that $d = q$ can write the answer; only $O_q > 0$ shows that the label names the effective direction. A concept can therefore fail ownership in two ways: its own write stays within the random and sham references, or the write clears them without holding its advantage over every competitor. Section 5.1 measures both. The Effusion case of the introduction illustrates the latter: the Effusion direction raises the Effusion answer by 0.190, above the random 95th percentile and the sham, and the Nodule direction raises the same answer by 0.262, so $O_{\text{Effusion}} = -0.071$. Uncertainty over patients comes from 2,000 bootstrap draws of test rows, with the maximum in Equation 4 recomputed inside every draw. The 30 contrasts $W_{q,q} - W_{q,d}$ (six questions, five competitors each) receive simultaneous 95% bounds from a centred studentised max- T bootstrap, which hold jointly across all 30 (Appendix E).

3.4 Graded outcomes

The readable and answerable grades are computed on the 400 calibration rows, which carry no write. The write matrix and every ownership quantity are computed on the 600 test rows, which share no patient with the calibration rows.

- **Readable:** the one-sided 95% lower bound of S on the calibration rows is above zero, with at least ten known positives and ten known negatives among those 400 rows.
- **Answerable:** the *clean answer*, $P(\text{yes})$ with no write, separates the concept’s labels: the one-sided 95% lower bound of its AUROC on the same calibration rows is above 0.5, under the same ten-and-ten support rule.
- **Owned:** on the test rows, the write meets the *steering reference* ($W_{q,q} > 0$, above the 95th percentile of the 119 random writes, and above the sham’s absolute effect) and every simultaneous lower bound of $W_{q,q} - W_{q,d}$ is positive. A cell with any simultaneous upper bound below zero has a *stronger competitor*; the rest are *unresolved*.

A checkpoint–concept pair on one dataset is a *cell*, and a checkpoint–dataset pair is a *block*. A cell graded owned is said to be owned, and the checkpoint is said to own the concept; an alternative direction is owned in a cell when its write meets the same rule for that concept. A block contributes to the answerable and owned counts only when its write matrix and its calibration pass both completed. The grades are defined for one direction construction, one primary dose with a five-point sweep, six templates, pooled consumed-token features, and two loci, the consumed block and the connector output. Before a block is scored, seven acceptance checks on 16 held-out preflight rows confirm, at both loci, that scoring is deterministic, that $\alpha = 0$ changes nothing, that the write reaches the answer and changes nothing outside the consumed tokens, and that every template maps its answers correctly without an image (Appendix A.3).

4 Models, Datasets, and Scale

Checkpoints. We evaluate 25 instruction-tuned checkpoints from twelve families, 3B to 72B (Table 2). Six families are general-purpose: Qwen2.5-VL (Bai et al., 2025b), Qwen3-VL (Bai et al., 2025a), InternVL3.5 (Wang et al., 2025), Gemma 3 (Gemma Team, 2025), LLaVA-1.5 (Liu et al., 2024), and Llama 3.2 Vision (Meta, 2024). Six are medical fine-tunes: MedGemma (Selligren et al., 2025), Lingshu (LASA Team et al., 2025), LLaVA-Med 1.5 (Li et al., 2023), CheXagent (Chen et al., 2024b), HuatuoGPT-Vision (Chen et al., 2024a), and LLaVA-Rad (Zambrano Chaves et al., 2025). They span six vision-tower designs and three consumer designs: token insertion after a merger, token insertion after a projector, and cross-attention. Three towers are trained on chest radiographs, those of CheXagent-2-3B, CheXagent-8B, and LLaVA-Rad-7B (Appendix A.2). Every checkpoint runs in bf16 with its native processor.

Datasets and cohorts. NIH ChestX-ray14 (Wang et al., 2017) and CheXpert Plus (Chambon et al., 2024) provide frontal chest radiographs with report-derived labels, the CheXpert labels coming from the CheXpert labeler (Irvin et al., 2019). We evaluate six findings per dataset (NIH: Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, Nodule; CheXpert: Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Consolidation, Edema). COCO (Lin et al., 2014) provides the natural-image control, with binary presence labels for six object categories (person, dog, car, chair, bottle, bicycle). Seeded hashes of patient or image identity select one fixed cohort per dataset, shared by every checkpoint: 20,000 training rows (18,212 on NIH) fit the probes and the answer direction, 400 calibration rows carry the readable and answerable grades, 600 test rows carry every write, and 16 preflight rows carry the acceptance checks. No patient and no image appears in two roles of a dataset (Appendix F.5).

Questions and scale. Each concept is asked as a yes/no question in six templates that cross two wordings (*is there* versus *does the image show*) with three answer interfaces (yes/no, A-present, B-present); the *is/yes-no* template gives the primary grade (Appendix D). The primary dose is $\alpha = +0.25$, and a sweep adds $\alpha \in \{-0.5, -0.25, -0.1, +0.1, +0.5\}$. Each block scores the write matrix, the sham, and the 119 random directions on the 600 test rows. The campaign comprises 75 blocks, 450 scored concept cells, about 30 million scored answers, and roughly 3,000 A100-hours; 74 of the 75 blocks carry every planned analysis (Appendix A.4).

5 Results

5.1 Reading is not writing

On both chest datasets, reading and answering far outpace writing (Table 1, Figure 2a–b). Clinical concepts are readable in 178 of 300 cells (59%; median selectivity 0.14), and 199 (66%) are answerable. Yet only 38 cells (13%) are owned, and reading and answering together barely change this: 28 of the 141 readable and answerable cells are owned (20%), against 10 of the remaining 159 (6%).

Ownership on chest radiographs breaks at the write itself. 244 of 300 label directions fail the random and sham reference for their own question, and these cells carry 187 of the 196 stronger-competitor verdicts (95%). Once an own write clears the reference, 38 of 56 cells are owned; on COCO, 117 of 150 writes clear it and 113 are owned. On chest radiographs, most label directions therefore scarcely move the answers they name, while the directions of other findings often move those answers more.

The Effusion case reproduces on 600 new patients (Table 15). The Effusion write raises its answer by 0.190, second among the 120 compared effects, the Nodule write raises it by 0.262, and $O_{\text{Effusion}} = -0.071 [-0.080, -0.062]$, matching the $-0.057 [-0.069, -0.045]$ of the two-model pilot study that motivated the evaluation (Appendix B). Across the grid, Effusion is owned in 7 of 50 chest blocks, and in 6 of these $O_{\text{Effusion}} \leq 0.03$.

Table 1: **Reading, answering, and writing per checkpoint.** Per dataset: mean probe selectivity S (READ), mean clean-answer AUROC (ANS), and mean ownership O_q (OWN), each superscripted with the number of the six concepts that receive the grade; then the owned share of clinical and COCO cells and their ratio, by which rows are sorted. Shading darkens with magnitude (blue for negative OWN); “–” marks an undefined ratio.

Checkpoint	NIH ChestX-ray14			CheXpert Plus			COCO (control)			Owned share		
	READ	ANS	OWN	READ	ANS	OWN	READ	ANS	OWN	clin.	COCO	ratio
Lingshu 32B	0.08 ³	0.78 ⁶	-0.02 ²	0.20 ⁴	0.87 ⁵	+0.04 ⁵	0.03 ¹	0.95 ⁵	+0.28 ⁶	58%	100%	58%
CheXagent-2-3B	0.13 ⁵	0.86 ⁶	+0.00 ²	0.21 ⁵	0.95 ⁵	+0.01 ⁴	0.13 ⁴	0.81 ⁵	+0.03 ²	50%	33%	150%
Qwen2.5-VL-72B	0.07 ²	0.62 ⁴	-0.03 ³	0.20 ⁴	0.71 ⁴	-0.09 ¹	0.01 ¹	0.99 ⁵	+0.66 ⁶	33%	100%	33%
Lingshu 7B	0.08 ³	0.76 ⁶	+0.02 ¹	0.21 ⁴	0.86 ⁵	+0.04 ³	0.02 ¹	0.97 ⁵	+0.68 ⁶	33%	100%	33%
Qwen3-VL-4B	0.08 ³	0.71 ⁶	-0.11 ⁰	0.20 ⁴	0.74 ⁴	-0.01 ²	0.04 ⁴	0.99 ⁵	+0.34 ⁶	17%	100%	17%
Qwen3-VL-32B	0.10 ²	0.70 ⁶	-0.02 ¹	0.17 ⁵	0.82 ⁵	-0.02 ¹	0.05 ⁴	0.99 ⁵	+0.15 ⁶	17%	100%	17%
InternVL3.5-38B	0.10 ⁵	0.78 ⁶	-0.00 ⁰	0.17 ⁴	0.86 ⁵	-0.00 ²	0.08 ⁴	0.99 ⁵	+0.01 ⁶	17%	100%	17%
Gemma 3 12B	0.07 ²	0.62 ⁵	-0.37 ¹	0.18 ⁴	0.52 ⁰	-0.38 ¹	0.07 ⁵	0.95 ⁵	+0.44 ⁶	17%	100%	17%
MedGemma 4B	0.12 ³	0.80 ⁶	-0.13 ⁰	0.21 ⁴	0.87 ⁵	-0.16 ²	0.06 ⁴	0.94 ⁵	+0.23 ⁶	17%	100%	17%
InternVL3.5-14B	0.07 ⁴	0.73 ⁶	-0.00 ¹	0.16 ⁴	0.87 ⁵	+0.00 ¹	0.12 ⁴	0.98 ⁵	+0.02 ⁵	17%	83%	20%
Qwen2.5-VL-32B	0.09 ³	0.63 ⁴	-0.06 ¹	0.16 ⁴	0.70 ⁴	-0.11 ⁰	0.02 ¹	0.99 ⁵	+0.38 ⁶	8%	100%	8%
InternVL3.5-8B	0.06 ²	0.73 ⁶	-0.01 ⁰	0.19 ⁴	0.86 ⁵	-0.00 ¹	0.12 ⁵	0.98 ⁵	+0.03 ⁶	8%	100%	8%
HuatuoGPT-Vision-7B	0.08 ³	0.71 ⁶	-0.03 ¹	0.19 ⁴	0.78 ⁴	-0.01 ⁰	0.07 ⁴	0.99 ⁵	+0.01 ⁴	8%	67%	12%
LLaVA-Med 1.5 7B	0.08 ³	0.53 ⁰	-0.01 ⁰	0.19 ⁴	0.59 ²	-0.01 ¹	0.07 ⁴	0.76 ⁵	+0.00 ¹	8%	17%	50%
CheXagent-8B	0.16 ⁵	0.73 ⁶	-0.00 ⁰	0.23 ⁵	0.88 ⁵	-0.02 ¹	0.05 ³	0.52 ¹	-0.03 ⁰	8%	0%	–
Qwen2.5-VL-3B	0.07 ²	0.58 ²	-0.08 ⁰	0.17 ⁴	0.57 ²	-0.07 ⁰	0.05 ⁴	0.97 ⁵	+0.52 ⁶	0%	100%	0%
Qwen2.5-VL-7B	0.08 ³	0.62 ³	-0.14 ⁰	0.20 ⁴	0.65 ²	-0.08 ⁰	0.02 ¹	0.98 ⁵	+0.67 ⁶	0%	100%	0%
Qwen3-VL-8B	0.10 ⁴	0.69 ⁶	-0.01 ⁰	0.17 ⁴	0.75 ⁴	-0.04 ⁰	0.04 ³	0.99 ⁵	+0.58 ⁶	0%	100%	0%
Gemma 3 27B	0.07 ²	0.60 ³	-0.03 ⁰	0.18 ⁴	0.55 ²	-0.01 ⁰	0.07 ⁵	0.96 ⁵	+0.31 ⁶	0%	100%	0%
Gemma 3 4B	0.07 ²	0.56 ²	-0.11 ⁰	0.18 ⁴	0.53 ¹	-0.27 ⁰	0.07 ⁵	0.94 ⁵	+0.24 ⁵	0%	83%	0%
LLaVA-1.5-7B	0.08 ²	0.51 ⁰	-0.04 ⁰	0.17 ⁴	0.52 ⁰	-0.03 ⁰	0.28 ⁵	0.97 ⁵	+0.03 ⁵	0%	83%	0%
MedGemma 27B	0.12 ³	0.80 ⁶	-0.14 ⁰	0.21 ⁴	0.87 ⁵	-0.06 ⁰	0.06 ⁴	0.97 ⁵	+0.12 ⁴	0%	67%	0%
LLaVA-1.5-13B	0.08 ²	0.53 ⁰	-0.02 ⁰	0.17 ⁴	0.51 ¹	-0.01 ⁰	0.28 ⁵	0.96 ⁵	+0.02 ²	0%	33%	0%
Llama 3.2 Vision 11B	0.08 ²	0.65 ⁴	-0.03 ⁰	0.16 ⁴	0.73 ⁴	-0.04 ⁰	0.12 ⁵	0.95 ⁵	-0.01 ¹	0%	17%	0%
LLaVA-Rad-7B	0.15 ⁴	0.71 ⁵	-0.01 ⁰	0.27 ⁵	0.87 ⁵	-0.02 ⁰	0.12 ⁴	0.50 ⁰	-0.03 ⁰	0%	0%	–
All cells	0.09	0.68	-0.06	0.19	0.74	-0.05	0.08	0.92	+0.23	13%	75%	17%
count / cells	74/150	110/150	13/150	104/150	89/150	25/150	90/150	116/150	113/150			

5.2 Where the chest write goes

Clinical writes are not selective (Figure 3). The written concept’s own question receives 24% of the positive answer change on NIH and 22% on CheXpert, against 79% on COCO. The write matrices show the same structure (Figure 5). The own write is small, with median $W_{q,q}$ of 0.010 on NIH and 0.020 on CheXpert against 0.127 on COCO, and the competing directions share what effect there is: the Atelectasis, Effusion, and Consolidation directions raise each other’s answers by similar amounts, and on the chest sets a row of W is flat or peaks off the diagonal, whereas on COCO the diagonal dominates. At every Qwen2.5-VL size a competing direction raises the Effusion answer more than the Effusion direction does (Cardiomegaly at 3B, Nodule at 7B and 32B, Atelectasis at 72B). Single images show the pattern (Figure 4): on a radiograph, the Effusion write raises the Effusion answer from 0.02 to 0.56 and the Nodule write raises it further, to 0.75, while on a natural image the dog write moves the dog answer from 0.00 to 1.00 and nothing else. The chest cells that are owned are also fragile: an effect floor of 0.05 on both $W_{q,q}$ and O_q keeps 18 of the 38 owned chest cells against 86 of the 113 on COCO, and ownership survives both probe refits in 4 of 13 owned NIH cells against 99 of 113 on COCO (Table 8).

5.3 The write mechanism is intact on natural objects

On COCO, the same procedure yields the opposite result (Table 1, right block). Object ownership holds in 113 of 150 cells (75%), 117 cells meet the steering reference, only 17 have a stronger com-

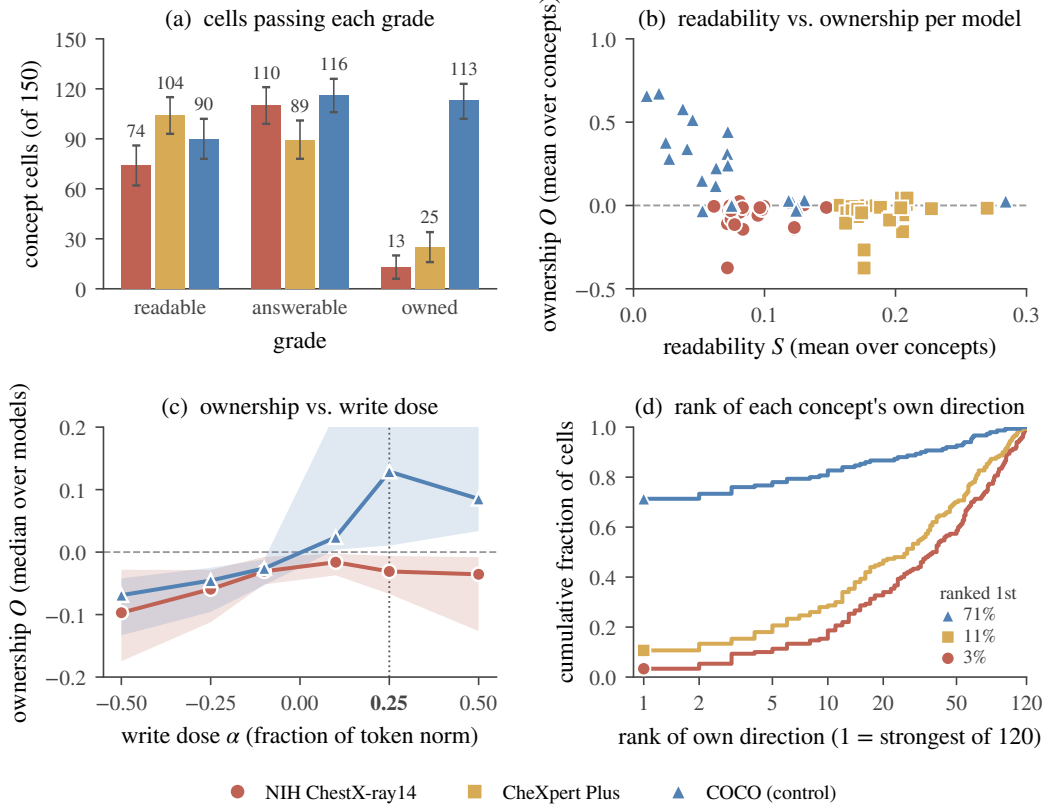


Figure 2: **Reading, answering, and writing across the grid.** (a) Readable, answerable, and owned cells of 150 per dataset, with descriptive 95% bootstrap intervals over cells. (b) Mean readability against mean ownership per checkpoint. (c) Median ownership and interquartile band over 25 blocks against the dose; the dotted line marks the graded dose $\alpha = +0.25$, and the COCO band is clipped at 0.2. (d) Cumulative share of cells by the rank of the concept write among the 120 compared effects.

petitor, and 19 of the 25 checkpoints own at least four of the six objects. Ownership also tracks the dose: on COCO, median ownership rises from -0.03 at $\alpha = -0.1$ to $+0.13$ at $\alpha = +0.25$ and stays positive at $\alpha = +0.5$ ($+0.09$), while on NIH it stays between -0.097 and -0.016 at every dose (Figure 2c). The write, direction construction, reference family, and decision rules thus work as intended.

The contrast survives two difficulty controls. Six small or fine-grained COCO categories (hand-bag, book, backpack, traffic light, knife, and cell phone) remain owned in 30 of 36 cells, against 34 of 36 for the easy six in the same 6 checkpoints (sign test $p = 0.25$), at a median clean-answer AUROC of 0.963 against 0.980. On the chest sets, ownership also stays rare where the model already answers the question well: at a clean-answer AUROC of at least 0.90, 11 of 37 chest cells are owned (30%) against 141 of 162 natural-image cells (87%), a difference of -0.573 [$-0.763, -0.388$] (Appendix A.11 states the category rule, the bins, and the bootstrap).

5.4 The reader decides ownership

Checkpoints that share a vision tower differ in what they own. Gemma 3 shares one SigLIP tower across its 4B, 12B, and 27B checkpoints, MedGemma one MedSigLIP tower across 4B and 27B, and LLaVA-1.5 one CLIP tower across 7B and 13B. Within each family, pooled features, fitted probes, and lifted write vectors are identical across sizes (Table 9), so readability is identical by construction and only the connector and language model that read the written representation differ. Steering a feature of the vision tower and letting the language model say what it sees treats that language model as a stable read-out of the tower (Ferrando et al., 2026); here the read-out is what varies. In 11 of

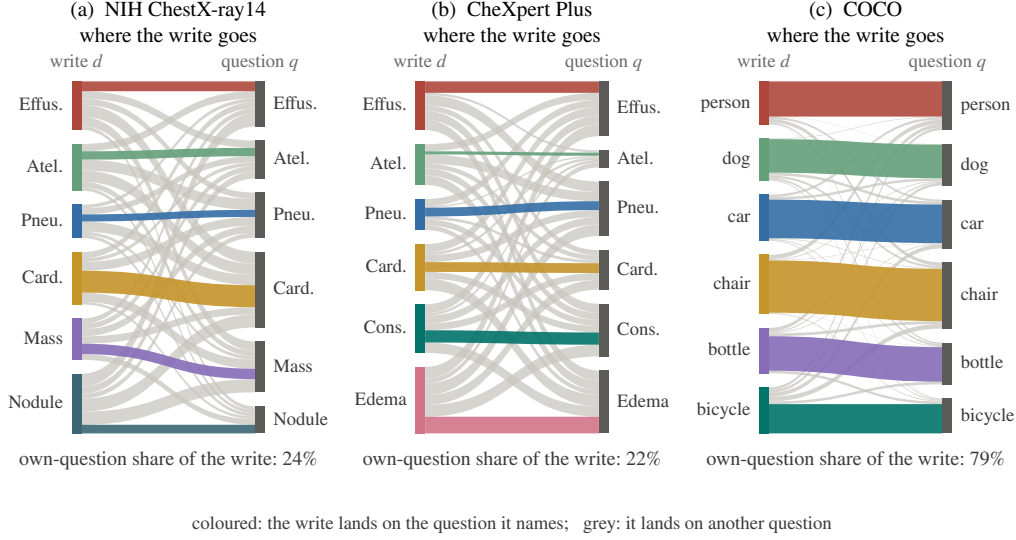


Figure 3: **Where the write goes.** Bands join each written direction d to the questions q it moves, with width proportional to $F_{q,d}$, the mean over checkpoints of $\max(W_{q,d}, 0)$. The percentage is the own-question share $\sum_d F_{d,d} / \sum_{q,d} F_{q,d}$.

the 12 shared-tower pair–dataset combinations, the paired patient-bootstrap interval for the between-reader difference ΔO_q excludes zero for five or six of the six concepts (Figure 6). On NIH Effusion, the identical write on the same 600 patients yields an O_q difference of -0.83 $[-0.85, -0.81]$ between Gemma 3 4B and 12B. In 12B, competing directions win the chest answers outright ($O_{\text{Effusion}} = -0.83$ on both chest sets), and LLaVA-1.5 7B and 13B own no chest concept yet own 5 and 2 COCO objects with the same write (Appendix A.8).

A better chest representation does not supply the missing reader. The three checkpoints whose towers are trained on chest radiographs read and answer the findings best in the grid (median probe AUROC 0.855 on NIH against 0.774 for the other NIH cells; median clean-answer AUROC 0.837 against 0.797 and 0.652 for medically tuned and general checkpoints), yet they own 7 of their 36 chest cells, against 15 of 72 for medically tuned checkpoints on general towers and 16 of 192 for general checkpoints. Scale raises chest ownership within some families, from none at Qwen2.5-VL 3B and 7B to three NIH concepts at 72B and from three to five CheXpert concepts across Lingshu 7B and 32B, but answering alone does not produce it: InternVL3.5 answers every NIH question at every size while its chest writes stay within 0.042 of zero (Figure 7). Ownership is therefore a property of the checkpoint that reads the representation, and a probe fitted to the tower alone cannot certify it.

5.5 Non-clinical attributes of the radiograph share the deficit

A non-clinical property of the same film is owned as rarely as a finding. We combine projection, sex, age, and the findings into one nine-direction family. The attribute probes read their labels at least as well as the clinical probes read the findings (median probe AUROC 0.896–0.999 against 0.808; Table 10), yet attribute cells are owned in 12 of 114 cases and finding cells in 23 of 228, the same rate. At a clean-answer AUROC of at least 0.90, attributes are owned in 0 of 13 cells and findings in 6 of 24. The deficit therefore follows the chest-radiograph representation and its reader rather than the concept’s clinical status.

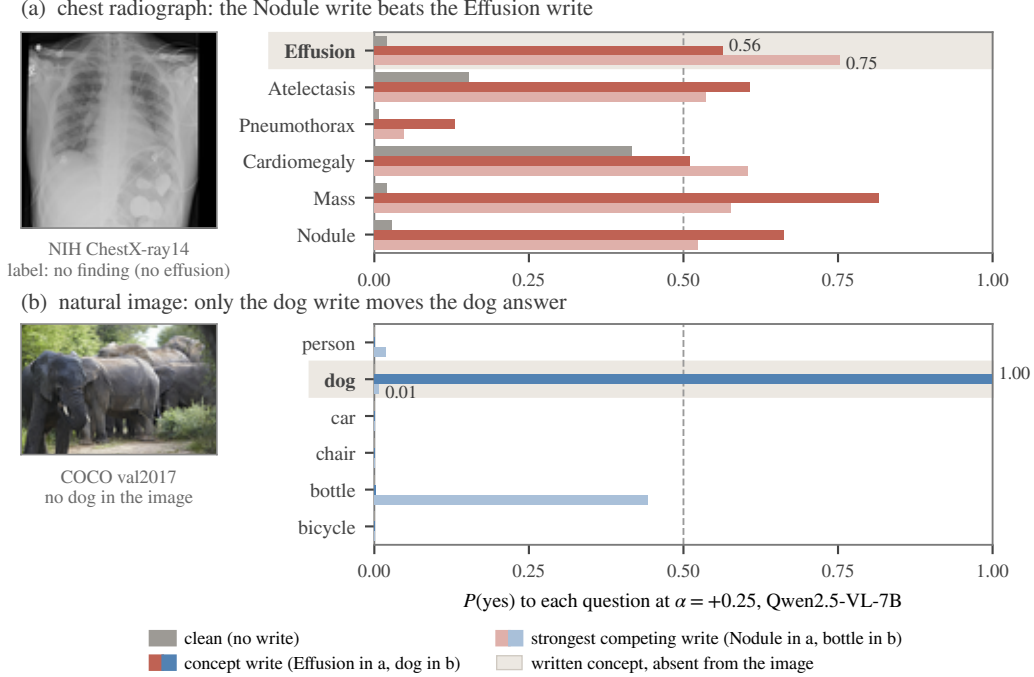


Figure 4: **Two images, six questions, three writes.** $P(\text{yes})$ per question for an NIH radiograph (top) and a COCO image (bottom), clean and under the concept write and the strongest competing write ($\alpha = +0.25$, Qwen2.5-VL-7B). The shaded row is the written concept, absent from the image; the dashed line marks $P(\text{yes}) = 0.5$.

5.6 The direction the reader responds to

For each question we fit the model’s own clean answer margin $\ell_{\text{yes}} - \ell_{\text{no}}$ on training rows from the same projected, train-scaled features by ridge regression, lift the fitted coefficients with Equation 2, and write the result on the test rows exactly as we write a label direction (Table 11). This *answer direction* a_q is a handle where the label direction is not. Across the 36 chest cells of 6 blocks, a_q is owned in 19 and the label direction in 9, and a_q remains owned in 76 of 95 chest (cell, template) pairs under the five held-out templates. The features that identify a concept differ from the features that control the output (Arad et al., 2025); here that separation holds against matched clinical competitors.

The answer direction reads the model, not the label. Over the 36 chest cells, its median AUROC against the report label is 0.66, against 0.80 for the probe, while against the model’s own clean answer it is 0.87, against 0.68 (Table 12). The two directions are far apart on chest radiographs, with a median raw model-space cosine of 0.18, and substantially aligned on natural images (0.59). The separation persists at four endpoints that fit neither direction (Table 13): the negated question, a forced choice against the strongest competitor in each of its two presentation orders, and the finding word’s log-probability in a report continuation. Across the 72 endpoint cells, the answer direction is owned in 30 and the label direction in 10. On chest radiographs, the direction the model answers along is not the direction the probe fits.

5.7 Controls and alternatives

The reference family behaves as designed: owned concept writes rank 1–6 among the 120 compared effects, and the concept write typically ranks in the top decile on the chest sets (Figure 2d). The deficit survives every alternative we tested; Appendix A.6 states the reference and rule of each of the 20 graded analyses, and Appendix A.7 gives the details.

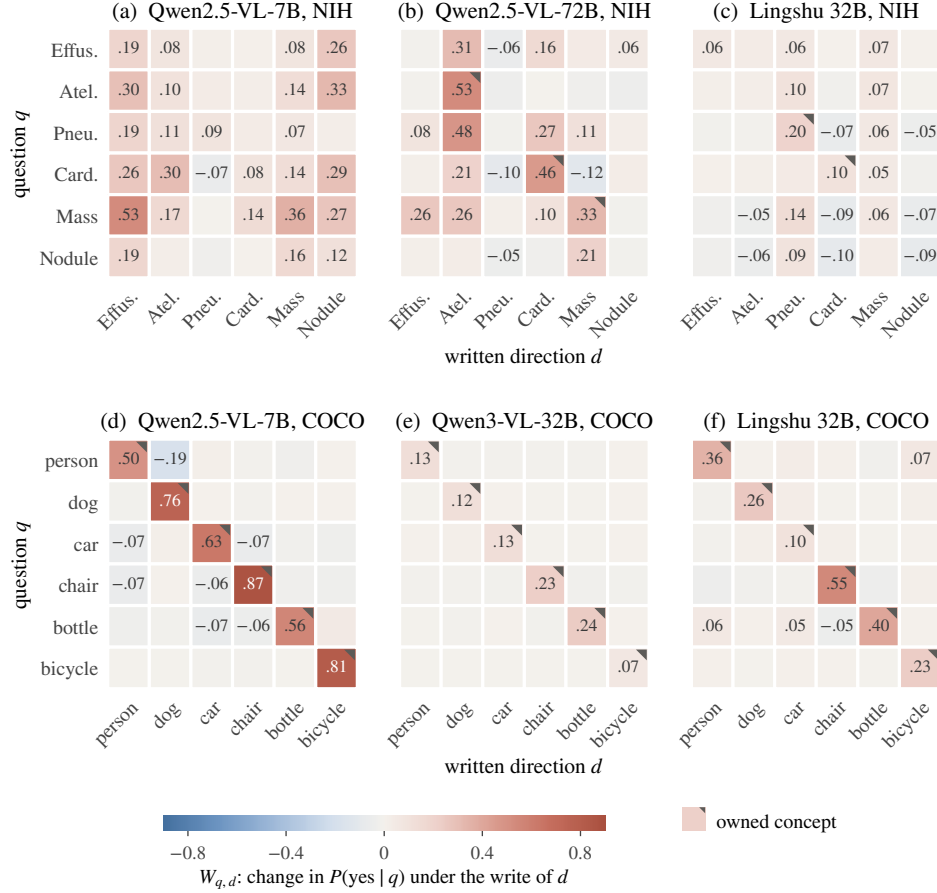


Figure 5: **Write matrices.** $W_{q,d}$, the mean change in $P(\text{yes})$ of question q (rows) under the write of direction d (columns), for three checkpoints on NIH (top) and three on COCO (bottom). A corner mark flags owned concepts.

Geometry, scale, and numerics. The strongest competitor matches the most similar direction or the most co-occurring label only at chance rate (Table 7). The logit-scale grade agrees with the probability-scale grade in 440 of 450 cells; restricting each cell to unsaturated rows preserves the owned grade in 36 of 38 owned chest cells; refitting the probes on resampled training patients changes O_q by a median of 0.02; and rescaling in fp32 at batch size one alters 1 of 108 grades (Appendix A.7).

Labels. Re-grading the fitted directions on 200 CheXpert films labelled by radiologist consensus (Irvin et al., 2019) reproduces the owned grade in 112 of 114 cells, and directions refitted on those labels are owned in 5 of 36 cells, against 12 for the report-label directions (Appendix A.7).

Directions. Four alternative constructions, the difference of class means, the Haufe pattern (the training covariance applied to the probe weights; Haufe et al., 2014), the normal orthogonalised against the other five normals, and the normal refitted on features residualised against the other five labels (Nicolson et al., 2025), are each owned in 6–16% of a chest dataset’s cells against 44–85% of COCO cells. Extending the competitor family to every additional dataset label with enough support retains 8 of the 11 owned cells, and writes concentrated on the tokens the probe reads are owned in 0 (softmax-weighted) and 5 (top quarter) of 36 chest cells, against 8 for the uniform write (Table 6). Concentrating the write where the probe reads does not create a clinical handle.

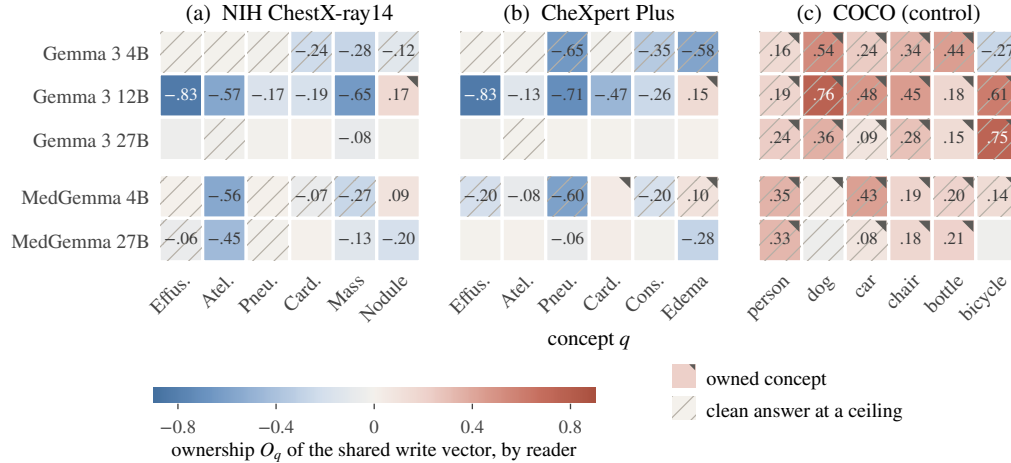


Figure 6: **Same write, different reader.** Ownership O_q per concept on NIH, CheXpert, and COCO for Gemma 3 4B/12B/27B and MedGemma 4B/27B, whose probes and write vectors are identical across sizes within each family. Corner marks flag owned cells and hatching marks ceiling cells.

6 Discussion

Readability belongs to the probe, ownership to the reader. Readability describes a classifier fitted to the representation; ownership describes how the model reading that representation responds to a write. Checkpoints sharing a vision tower have identical probes and write vectors, yet own different concepts, and the chest-trained towers read and answer findings best in the grid but own only 7 of 36 chest cells. Ownership therefore belongs on the scorecard of the checkpoint to be deployed rather than on that of the vision tower it contains.

On chest radiographs, the label direction and the answer direction come apart. On natural images, the label direction and the model’s answer direction are substantially aligned, and writing the label direction controls the answer. On chest radiographs the two are nearly orthogonal: the label direction barely moves its own answer while moving the other findings’ answers, and the answer direction is owned in about half of the chest cells. The flat rows of the chest write matrices fit a reader that responds to a signal shared across findings rather than to each finding’s own direction (Pahde et al., 2025); which signal that is remains open. A legible label need not mark the direction along which the reader chooses its answer, so concept control is a property of a direction and a reader together.

What a steering evaluation should report. Probe accuracy and answer accuracy, the two quantities most often reported for medical VLMs, are both high in this grid, yet ownership is 13%. Random and sham writes show that a direction reaches the answer; only directions fitted the same way for the competing findings, written at the same locus and dose in the same patients, show whether the direction belongs to its label. Studies that use probe normals as interventions should report the write matrix W , compare O_q across layers, doses, and models, and pair every clinical write with a natural-image control, where the same procedure produces selective control.

7 Conclusion

Reading a clinical concept from a vision-language model’s visual representation does not make its probe direction a handle on that concept. Across 25 checkpoints and two chest-radiograph datasets, concepts are readable in 59% and answerable in 66% of cells but owned in 13%, because the concept’s own direction barely moves its answer while the directions fitted for the other findings move it as much

or more. Under the same procedure 75% of natural-object cells are owned, also at matched answer ability. Checkpoints that write identical vectors own different concepts, and the model’s own answer direction, nearly orthogonal to the probe direction on chest radiographs, provides the handle that the label direction lacks. Readability is a property of a classifier fitted to the representation; control is a property of a direction and the model that reads it.

8 Limitations

The grades describe linear writes at two loci, the consumed block and the connector output, at one primary dose with a five-point sweep; deeper language-model layers and nonlinear edits may behave differently. Each dataset contributes six concepts, and the chest labels are report-derived, with radiologist labels available for a CheXpert validation subset. The grade needs a discrete answer. MAIRA-2 (Bannur et al., 2024) emits reports rather than answers and fails the image-free mapping check on every template (Appendix A.3), and the discrete-diffusion report generator of Chen et al. (2026) has the same shape, so both fall outside the evaluation.

Ownership is a verdict on the write comparison rather than a predictor of effect size. On the four semantic endpoints, adding the ownership contrast to write magnitude, probe selectivity, and clean-answer AUROC adds no held-out predictive value beyond a permutation null ($p = 0.26$ and $p = 0.26$ under the concept and block splits; Appendix A.10). The answer direction is fitted on training rows to the model’s own clean answers and written on test rows and held-out templates, so it is a handle on the model’s answer. Its median AUROC against the report label is 0.66, below the probe’s 0.80, and answer control and clinical fidelity remain distinct objectives.

Reproducibility statement

The registered protocol file fixes the loci, projection, probe settings, dose grid, reference family, templates, modules, gates, and decision rules. The release contains the evaluation code and adapters, the protocol file, the cohort manifests with their seeds, the fitted directions, the per-image outcomes of every module, every block’s scorecard and coverage file, the rendered prompts, and the pinned checkpoint revisions (Table 2); every table and figure in the paper is regenerated from those files by the scripts in the release. Appendix F states the consumed block of every family, the consumed-token mask, the projection and probe construction, the control sampling procedures, the cohort roles, and the protocol amendments with their dates.

Ethics statement

The study uses three public datasets under their published terms: NIH ChestX-ray14 (NIH Clinical Center), CheXpert Plus (Stanford AIMI, accessed through its data-use agreement), and COCO (annotations CC BY 4.0). All radiographs are de-identified by their providers; no protected health information was accessed, and no image or report is redistributed. The write test evaluates model behaviour under activation edits and is not a diagnostic tool; its grades describe what a probe direction does inside a model and carry no claim of clinical benefit.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations Workshop*, 2017.
- Dana Arad, Aaron Mueller, and Yonatan Belinkov. SAEs are good for steering—if you select the right features. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 10241–10259, 2025.

- Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, et al. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025a.
- Shuai Bai, Keqin Chen, Xuejing Liu, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025b.
- Shruthi Bannur, Kenza Bouzid, Daniel C. Castro, Anton Schwaighofer, Anja Thieme, Sam Bond-Taylor, Maximilian Ilse, Fernando Pérez-García, Valentina Salvatelli, Harshita Sharma, et al. MAIRA-2: Grounded radiology report generation. *arXiv preprint arXiv:2406.04449*, 2024.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1), 2022.
- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Pierre Chambon, Jean-Benoit Delbrouck, Thomas Sounack, Shih-Cheng Huang, Zhihong Chen, Maya Varma, Steven QH Truong, Chu The Chuong, and Curtis P. Langlotz. CheXpert Plus: Augmenting a large chest x-ray dataset with text radiology reports, patient demographics and additional image formats. *arXiv preprint arXiv:2405.19538*, 2024.
- Junying Chen, Chi Gui, Ruyi Ouyang, Anningzhe Gao, Shunian Chen, Guiming Hardy Chen, Xidong Wang, Ruifei Zhang, Zhenyang Cai, Ke Ji, et al. HuatuoGPT-Vision, towards injecting medical visual knowledge into multimodal LLMs at scale. *arXiv preprint arXiv:2406.19280*, 2024a.
- Lifeng Chen, Tianqi You, Hao Liu, Zhimin Bao, Jile Jiao, Xiao Han, Zhicai Ou, Tao Sun, Xiaofeng Mou, Xiaojie Jin, and Yi Xu. ECHO: Efficient chest x-ray report generation with one-step block diffusion. *arXiv preprint arXiv:2604.09450*, 2026.
- Zhihong Chen, Maya Varma, Justin Xu, Magdalini Paschali, Dave Van Veen, Andrew Johnston, Alaa Youssef, Louis Blankemeier, Christian Bluethgen, Stephan Altmayer, et al. A vision-language foundation model to enhance efficiency of chest X-ray interpretation. *arXiv preprint arXiv:2401.12208*, 2024b.
- Zehao Deng, Tianjie Ju, Zheng Wu, Liangbo He, Jun Lan, Huijia Zhu, Weiqiang Wang, and Zhuosheng Zhang. Causal probing for internal visual representations in multimodal large language models. *arXiv preprint arXiv:2605.05593*, 2026. Accepted at EMNLP 2026 Main.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021.
- Javier Ferrando, Enrique Lopez-Cuena, Pablo Agustin Martin-Torres, Daniel Hincos, Anna Arias-Duart, and Dario Garcia-Gasulla. Language models can explain visual features via steering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Conference on Causal Learning and Reasoning (CLear)*, pages 160–187, 2024.
- Gemma Team. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- Stefan Haufe, Frank Meinecke, Kai Görden, Sven Dähne, John-Dylan Haynes, Benjamin Blankertz, and Felix Bießmann. On the interpretation of weight vectors of linear models in multivariate neuroimaging. *NeuroImage*, 87:96–110, 2014.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 2733–2743, 2019.

- Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghgoo, Robyn Ball, Katie Shpanskaya, Jayne Seekins, David A. Mong, Safwan S. Halabi, Jesse K. Sandberg, Ricky Jones, David B. Larson, Curtis P. Langlotz, Bhavik N. Patel, Matthew P. Lungren, and Andrew Y. Ng. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 590–597, 2019.
- Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In *International Conference on Machine Learning (ICML)*, pages 2668–2677, 2018.
- LASA Team, Weiwen Xu, Hou Pong Chan, Long Li, Mahani Aljunied, Ruifeng Yuan, et al. Lingshu: A generalist foundation model for unified multimodal medical understanding and reasoning. *arXiv preprint arXiv:2506.07044*, 2025.
- Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision (ECCV)*, pages 740–755, 2014.
- Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2024.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *First Conference on Language Modeling (COLM)*, 2024.
- Meta. Llama 3.2-Vision model card. https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD_VISION.md, 2024.
- Angus Nicolson, Lisa Schut, J. Alison Noble, and Yarin Gal. Explaining explainability: Recommendations for effective use of concept activation vectors. *Transactions on Machine Learning Research*, 2025.
- Frederik Pahde, Maximilian Dreyer, Leander Weber, Moritz Weckbecker, Christopher J. Anders, Thomas Wiegand, Wojciech Samek, and Sebastian Lapuschkin. Navigating neural space: Revisiting concept activation vectors to overcome directional divergence. In *International Conference on Learning Representations (ICLR)*, 2025.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning (ICML)*, pages 39643–39666, 2024.
- Nikollette Pedersen, Regitze Syddendal, Veronika Cheplygina, and Théo Sourget. Look what the probes dragged in! real-world chest x-ray shortcuts in MedCLIP. *arXiv preprint arXiv:2608.12086*, 2026.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.

- Achyuta Rajaram, Sarah Schwettmann, Jacob Andreas, and Arthur Conmy. Line of sight: On linear representations in VLLMs. *arXiv preprint arXiv:2506.04706*, 2025.
- Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, 2020.
- Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 3363–3377, 2021.
- Andrew Sellergren, Sahar Kazemzadeh, Tiam Jaroensri, Atilla Kiraly, Madeleine Traverse, Timo Kohlberger, Shawn Xu, Fayaz Jamil, Cían Hughes, Charles Lau, et al. MedGemma technical report. *arXiv preprint arXiv:2507.05201*, 2025.
- Quan Sun, Yuxin Fang, Ledell Wu, Xinlong Wang, and Yue Cao. EVA-CLIP: Improved training techniques for CLIP at scale. *arXiv preprint arXiv:2303.15389*, 2023.
- Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *arXiv preprint arXiv:2502.14786*, 2025.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.
- Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, et al. InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.
- Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 2097–2106, 2017.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders. In *International Conference on Machine Learning (ICML)*, pages 67035–67080, 2025.
- Juan Manuel Zambrano Chaves, Shih-Cheng Huang, Yanbo Xu, Hanwen Xu, Naoto Usuyama, Sheng Zhang, Fei Wang, Yujia Xie, Mahmoud Khademi, Ziyi Yang, et al. A clinically accessible small multimodal radiology model and evaluation metric for chest X-ray findings. *Nature Communications*, 16(1):3108, 2025.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- Sheng Zhang, Yanbo Xu, Naoto Usuyama, Hanwen Xu, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, et al. A multimodal biomedical foundation model trained from fifteen million image–text pairs. *NEJM AI*, 2(1), 2025.
- Xun Zhu, Fanbin Mo, Xi Chen, Kaili Zheng, Shaoshuai Yang, Yiming Shi, Jian Gao, Miao Li, and Ji Wu. Lost in the hype: Revealing and dissecting the performance degradation of medical multimodal large language models in image classification. *arXiv preprint arXiv:2604.08333*, 2026.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A Campaign details

This appendix lists the checkpoints and their consumed loci, the acceptance decisions, the coverage of the campaign, and the per-concept results of every completed block.

A.1 Labels, registration, and artifact

Labels. We use each dataset’s supplied report-derived labels: NIH ChestX-ray14 finding labels (Wang et al., 2017), the CheXpert labeler output (Irvin et al., 2019) distributed with CheXpert Plus (Chambon et al., 2024), and COCO instance annotations (Lin et al., 2014). Rows are positive for label 1, negative for label 0, and *unknown* when the labeler output is uncertain or blank, which is the uncertainty convention of the CheXpert labeler (Irvin et al., 2019). We exclude unknown rows from probe fitting, selectivity, and answer AUROC. We write and score them like all other rows. Fixed manifests drawn with recorded seeds define the cohorts, one manifest per dataset for every block. No patient and no image appears in two roles of a dataset. Every cohort is disjoint from every pilot-study cohort. Three cohorts are scored: the training cohort for fitting the scaler, probes, and answer direction; the 400-row calibration cohort for the readable and answerable grades; and the 600-row test cohort for every write. A second CheXpert cohort contains 200 frontal films from the CheXpert validation set, one per patient, with the radiologists’ consensus labels (Irvin et al., 2019). This cohort has zero patient overlap with any campaign cohort.

Registration. We froze the protocol before scoring any grid block. The loci, projection and probe settings, dose, reference family, six templates, module set, seven gates, and three decision rules match the registered protocol file. We registered the pilot study’s crossover and confirmation analyses before running them (Appendix B). After completing the grid, we added the leaderboard, write-flow share, and per-image examples as descriptive summaries of these outputs.

Artifact. The release includes evaluation code, adapters, the registered protocol file, cohort manifests with their seeds, and fitted directions. It provides per-image outcomes for every module (write, dose, refit, locus, and template rows), scorecards and coverage files for every block, and rendered prompts. Each new checkpoint requires one adapter that identifies its input block and passes the gates.

A.2 Grid and consumed loci

The towers include CLIP (Radford et al., 2021) in the LLaVA families, SigLIP (Zhai et al., 2023) in Gemma 3, and a SigLIP 2 design (Tschannen et al., 2025) in Qwen3-VL; MedGemma’s MedSigLIP tower is a medically tuned SigLIP released with that model (Sellersgren et al., 2025). The three towers trained on chest radiographs are the XraySigLIP tower of CheXagent-2-3B (Chen et al., 2024b), the EVA-CLIP tower (Sun et al., 2023) of CheXagent-8B, and the BiomedCLIP-CXR tower of LLaVA-Rad-7B (Zhang et al., 2025; Zambrano Chaves et al., 2025); HuatuoGPT-Vision-7B tunes on medical data over the same stock CLIP tower as LLaVA-1.5. Consumed-token counts range from 128 to 4,096 per image, and each family uses a fixed image resolution. The Llama 3.2 Vision projector also reads five intermediate local-layer outputs that bypass the final global block. LLaVA computes the final CLIP block but consumes the penultimate block. At every checkpoint, Gate (C) verifies that the write reaches the consumer and the answer logits and changes nothing outside the consumed tokens.

A.3 Acceptance decisions

Seven checks run on the 16 preflight rows at both loci before a block is scored: (A) bitwise determinism; (B) identity at $\alpha = 0$; (C) reach, under which the write must change the consumer’s input and answer logits without changing anything outside the consumed tokens; (D) batched-versus-single agreement of the answer logits within 0.25; (E) image-free semantic mapping of every template; (F) throughput; and (G) fp32 recomputation of the answer logits. Deviations in (D) are declared and

Table 2: **Checkpoints and families.** Per checkpoint: vision tower, consumed tokens per image, input size, and eligible templates. Bold family rows give mean selectivity and clean-answer AUROC over clinical cells and the owned clinical and COCO cells, shaded by owned share.

Checkpoint	Vision tower	Tokens	Image	Templates	$\bar{S}$	AUROC	owned cells	
							clinical	COCO
Qwen2.5-VL (Bai et al., 2025b)					0.132	0.635	5/48	24/24
Qwen2.5-VL-3B	native ViT 675M	576→144	336 ²	IYWYIAWA				
Qwen2.5-VL-7B	native ViT 675M	576→144	336 ²	all				
Qwen2.5-VL-32B	native ViT 675M	576→144	336 ²	all				
Qwen2.5-VL-72B	native ViT 675M	576→144	336 ²	all				
Qwen3-VL (Bai et al., 2025a)					0.137	0.735	4/36	18/18
Qwen3-VL-4B	SigLIP2-based	≤1024	≤ 512 ²	all				
Qwen3-VL-8B	SigLIP2-based	≤1024	≤ 512 ²	all				
Qwen3-VL-32B	SigLIP2-based	≤1024	≤ 512 ²	all				
InternVL3.5 (Wang et al., 2025)					0.125	0.807	5/36	17/18
InternVL3.5-8B	InternViT-300M	1024	448 ²	all				
InternVL3.5-14B	InternViT-300M	1024	448 ²	all				
InternVL3.5-38B	InternViT-6B	1024	448 ²	all				
Gemma 3 (Gemma Team, 2025)					0.124	0.564	2/36	17/18
Gemma 3 4B	SigLIP-So400m	4096	896 ²	all				
Gemma 3 12B	SigLIP-So400m	4096	896 ²	all				
Gemma 3 27B	SigLIP-So400m	4096	896 ²	all				
MedGemma (medical) (Sellersgren et al., 2025)					0.164	0.835	2/24	10/12
MedGemma 4B	MedSigLIP	4096	896 ²	all				
MedGemma 27B	MedSigLIP	4096	896 ²	all				
Lingshu (medical) (LASA Team et al., 2025)					0.142	0.820	11/24	12/12
Lingshu 7B	Qwen2.5-VL ViT	576→144	336 ²	all				
Lingshu 32B	Qwen2.5-VL ViT	576→144	336 ²	all				
LLaVA-1.5 (Liu et al., 2024)					0.124	0.516	0/24	7/12
LLaVA-1.5-7B	CLIP ViT-L/14-336	576	336 ²	IYWY				
LLaVA-1.5-13B	CLIP ViT-L/14-336	576	336 ²	IYWYIAWA				
LLaVA-Med (medical) (Li et al., 2023)					0.136	0.560	1/12	1/6
LLaVA-Med 1.5 7B	CLIP ViT-L/14-336	576	336 ²	IYWY				
Llama 3.2 Vision (Meta, 2024)					0.119	0.688	0/12	1/6
Llama 3.2 Vision 11B	ViT-H/14	1601	560 ²	IBWB				
CheXagent (medical) (Chen et al., 2024b)					0.180	0.857	7/24	2/12
CheXagent-2-3B	XraySigLIP ViT-L/16	1024	512 ²	IYWYIAWA				
CheXagent-8B	EVA-CLIP ViT 1.4B	1025→128	448 ²	IYWYIAWA				
LLaVA-Rad (medical) (Zambrano Chaves et al., 2025)					0.209	0.793	0/12	0/6
LLaVA-Rad-7B	BiomedCLIP-CXR-518	1369	518 ²	IYWY				
HuatuoGPT-Vision (medical) (Chen et al., 2024a)					0.136	0.746	1/12	4/6
HuatuoGPT-Vision-7B	CLIP ViT-L/14-336	576	336 ²	all				

neutralised by scoring every condition of a question in one fixed batch composition. Templates that fail (E) are ineligible; a checkpoint whose yes/no template fails is graded on its eligible B-positive template (IB for Llama 3.2 Vision), and its scorecards record the template used and every declared deviation.

All 75 completed blocks pass checks (A), (B), (C), and (G): determinism error is zero, change at $\alpha = 0$ is zero, and fp32-versus-model logit differences are below 0.13. In check (D), the Qwen3-VL, Gemma 3, MedGemma, InternVL, and Llama families exceed the 0.25-logit tolerance (up to 1.84 logits for Gemma 3 12B). We record these as declared deviations. Fixed batch composition neutralises them. Check (E) fails across all datasets for the A/B templates for LLaVA-1.5-7B and LLaVA-Med, the B-mapped templates for Qwen2.5-VL-3B and LLaVA-1.5-13B, and the yes/no and A-mapped templates for Llama 3.2 Vision 11B. We record affected cells as ineligible. Check (E) fails for every template of MAIRA-2 (Bannur et al., 2024) on NIH: the model answers “No” to every finding on all 16 preflight images and “A” to every forced choice under both letter mappings, so no

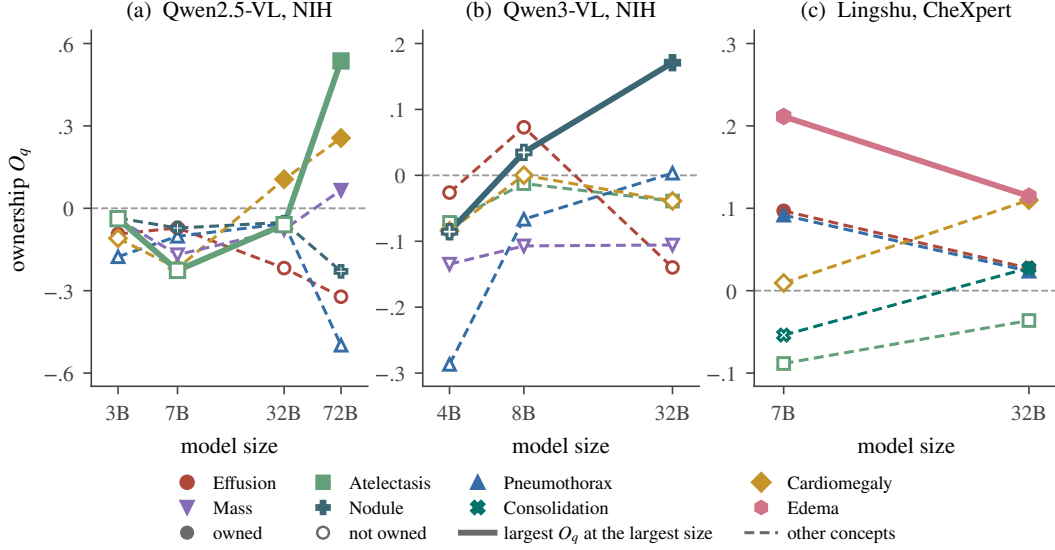


Figure 7: **Size ladders.** Ownership O_q per concept across sizes for Qwen2.5-VL and Qwen3-VL on NIH and Lingshu on CheXpert. Solid lines mark the concept with the largest ownership at the largest size (Atelectasis, Nodule, Edema), filled markers owned cells; vertical scales differ by panel.

template carries a graded answer and the block produces no outcomes. Two Gemma 3 checkpoints answer “yes” to most chest questions; Gemma 3 4B does so on 94–97% of rows (clean $P(\text{yes}) > 0.99$). The answerability grade records this behaviour. Three checkpoints (LLaVA-1.5 7B/13B, LLaVA-Med) answer at chance.

A.4 Coverage and cost

Every checkpoint has a completed core write matrix and calibration pass on NIH ChestX-ray14, CheXpert Plus, and COCO. The campaign targets seven modules per block: the write matrix, the calibration pass, the five additional templates, the dose sweep, the two probe refits, the connector locus, and the connector-locus calibration. 74 of 75 blocks carry all seven modules. A probe-graded block whose write matrix is ineligible or incomplete contributes readability alone. Each block’s scorecard also reports the dose range of O_q over $\alpha \in [-0.5, 0.5]$, its standard deviation over the three fit seeds, ownership at the connector locus, and the wording and mapping contrasts of O_q across the six templates. The partial block is Qwen2.5-VL-72B on CheXpert Plus, which lacks the additional templates and the connector locus. The campaign used about 3,000 A100-hours; a chest block of a 7B checkpoint takes about ten.

A.5 Ownership of every concept

Tables 3–5 report the ownership contrast O_q of every concept cell, one table per dataset, with the checkpoints in the order of Table 1. Their cells are shaded by magnitude at 0.02, 0.10, and 0.25, terracotta for positive and blue for negative ownership.

NIH ChestX-ray14. 13 of the 150 cells are owned, a stronger competitor is identified in 101, and the contrast is positive in 25% of cells. The owned cells concentrate in a few rows: Qwen2.5-VL-72B owns Atelectasis ($O_q = 0.54$), Cardiomegaly, and Mass, while in Gemma 3 12B the competing directions win most answers, down to $O_{\text{Effusion}} = -0.83$.

CheXpert Plus. 25 of the 150 cells are owned, 95 have a stronger competitor, and the contrast is positive in 29% of cells. Lingshu contributes the largest share of the owned cells, with three concepts at 7B and five at 32B.

Table 3: **Ownership of every concept, NIH ChestX-ray14.** O_q at $\alpha = +0.25$: bold is owned, $\dagger$ a stronger competitor, $\ddagger$ unresolved; the last row counts owned cells. Shading darkens with $|O_q|$ (terracotta positive, blue negative).

Checkpoint	Effusion	Atelectasis	Pneumothorax	Cardiomegaly	Mass	Nodule
Lingshu 32B	-0.01	-0.08	+0.14	+0.05	-0.08	-0.18
CheXagent-2-3B	-0.00 $\ddagger$	+0.04	-0.00 $\dagger$	-0.01	+0.01	-0.01
Qwen2.5-VL-72B	-0.32	+0.54	-0.50	+0.26	+0.07	-0.23
Lingshu 7B	+0.02	+0.01	+0.01	+0.17	-0.09	+0.02
Qwen3-VL-4B	-0.03 $\dagger$	-0.07	-0.29	-0.08	-0.13	-0.09
Qwen3-VL-32B	-0.14	-0.04	+0.00 $\ddagger$	-0.04	-0.11	+0.17
InternVL3.5-38B	+0.00 $\ddagger$	-0.02	-0.00	-0.03	+0.02	+0.00
Gemma 3 12B	-0.83	-0.57	-0.17	-0.19	-0.65	+0.17
MedGemma 4B	+0.01	-0.56	+0.01	-0.07	-0.27	+0.09
InternVL3.5-14B	+0.01	-0.01	-0.00	-0.00	+0.00	-0.01
Qwen2.5-VL-32B	-0.22	-0.06	-0.06	+0.11	-0.08	-0.05
InternVL3.5-8B	+0.00	-0.02	-0.00	+0.00	-0.01	-0.01
HuatuoGPT-Vision-7B	-0.04	-0.03	-0.03	-0.08	-0.04	+0.01
LLaVA-Med 1.5 7B	-0.00	-0.00	-0.04	-0.01	-0.01	-0.02
CheXagent-8B	+0.01	-0.01	-0.00	-0.00	+0.00	-0.01
Qwen2.5-VL-3B	-0.09	-0.04	-0.18	-0.11	-0.04	-0.04
Qwen2.5-VL-7B	-0.07 $\dagger$	-0.23	-0.10	-0.22	-0.17	-0.07
Qwen3-VL-8B	+0.07	-0.01	-0.07	+0.00	-0.11	+0.03
Gemma 3 27B	-0.04	-0.03	-0.01	+0.00 $\ddagger$	-0.08	-0.02
Gemma 3 4B	-0.00	-0.00	-0.01	-0.24	-0.28	-0.12
LLaVA-1.5-7B	-0.03	-0.01	-0.08	-0.13	+0.02	-0.02
MedGemma 27B	-0.06	-0.45	-0.01	+0.00	-0.13	-0.20
LLaVA-1.5-13B	-0.01	-0.05	-0.04	-0.03	+0.01	-0.02
Llama 3.2 Vision 11B	-0.04	-0.07	-0.05	+0.02	-0.06	-0.01
LLaVA-Rad-7B	-0.02	-0.01	-0.00	+0.00	-0.01	-0.03
Owned	1	2	1	4	2	3

COCO. 113 of the 150 cells are owned and 17 have a stronger competitor, and 19 of the 25 checkpoints own at least four of the six objects. The rows without owned objects are CheXagent-8B and LLaVA-Rad-7B, whose clean answers to the COCO questions are near chance (Table 1).

Table 4: Ownership of every concept, CheXpert Plus. As Table 3.

Checkpoint	Effusion	Atelectasis	Pneumothorax	Cardiomegaly	Consolidation	Edema
Lingshu 32B	+0.03	-0.04 [†]	+0.02	+0.11	+0.03	+0.11
CheXagent-2-3B	+0.03	-0.02	+0.01	+0.02	+0.01	-0.00 [‡]
Qwen2.5-VL-72B	-0.12	+0.09	+0.03	-0.11	-0.29	-0.13
Lingshu 7B	+0.10	-0.09	+0.09	+0.01 [‡]	-0.05	+0.21
Qwen3-VL-4B	-0.01 [†]	-0.02	-0.23	-0.01	+0.09	+0.09
Qwen3-VL-32B	-0.04	+0.00 [‡]	+0.00	-0.03	-0.12	+0.05
InternVL3.5-38B	+0.02	-0.01	-0.00	-0.00 [‡]	+0.01	-0.02
Gemma 3 12B	-0.83	-0.13	-0.71	-0.47	-0.26	+0.15
MedGemma 4B	-0.20	-0.08	-0.60	+0.04	-0.20	+0.10
InternVL3.5-14B	+0.01	-0.00	+0.00	-0.01	+0.01	-0.00
Qwen2.5-VL-32B	-0.07	-0.19	-0.12	-0.19	-0.05	-0.02
InternVL3.5-8B	-0.01	-0.01	+0.00	-0.00	+0.01	-0.00
HuatuoGPT-Vision-7B	-0.02	-0.02	-0.00	+0.01	-0.03	-0.01
LLaVA-Med 1.5 7B	+0.00	-0.00	-0.02	-0.03	-0.00 [†]	-0.02
CheXagent-8B	-0.02	-0.02	-0.02	-0.07	+0.04	-0.02
Qwen2.5-VL-3B	-0.18	-0.02	-0.04	-0.05	-0.03	-0.08
Qwen2.5-VL-7B	+0.01	-0.05	-0.10	-0.00	-0.15	-0.18
Qwen3-VL-8B	-0.07 [†]	-0.00 [†]	-0.10	-0.00	-0.04 [†]	-0.06
Gemma 3 27B	-0.03	-0.00	-0.01	-0.00	-0.03	+0.00 [‡]
Gemma 3 4B	-0.00	-0.00	-0.65	-0.02	-0.35	-0.58
LLaVA-1.5-7B	+0.00	-0.09	-0.10	+0.03	-0.03	-0.00
MedGemma 27B	-0.01	+0.00	-0.06	+0.03	-0.02	-0.28
LLaVA-1.5-13B	-0.00	-0.03	-0.01	+0.00	+0.01	-0.01
Llama 3.2 Vision 11B	-0.01	-0.05	-0.08	-0.07	-0.05	-0.01
LLaVA-Rad-7B	-0.04	-0.01	+0.00	-0.07	+0.02	-0.00
Owned	6	0	5	3	5	6

Table 5: **Ownership of every concept, COCO.** As Table 3.

Checkpoint	person	dog	car	chair	bottle	bicycle
Lingshu 32B	+0.29	+0.23	+0.08	+0.53	+0.34	+0.22
CheXagent-2-3B	+0.07	+0.07	+0.03	-0.01	+0.04	+0.00
Qwen2.5-VL-72B	+0.66	+0.44	+0.95	+0.89	+0.62	+0.40
Lingshu 7B	+0.39	+0.74	+0.70	+0.76	+0.64	+0.85
Qwen3-VL-4B	+0.46	+0.13	+0.33	+0.71	+0.34	+0.07
Qwen3-VL-32B	+0.13	+0.12	+0.11	+0.23	+0.24	+0.07
InternVL3.5-38B	+0.01	+0.00	+0.01	+0.02	+0.01	+0.01
Gemma 3 12B	+0.19	+0.76	+0.48	+0.45	+0.18	+0.61
MedGemma 4B	+0.35	+0.03	+0.43	+0.19	+0.20	+0.14
InternVL3.5-14B	+0.02	+0.01	+0.01	+0.04	+0.03	+0.00 [‡]
Qwen2.5-VL-32B	+0.44	+0.12	+0.27	+0.83	+0.41	+0.19
InternVL3.5-8B	+0.03	+0.01	+0.04	+0.04	+0.04	+0.02
HuatuoGPT-Vision-7B	+0.02	+0.00 [‡]	+0.00	+0.03	+0.03	+0.00 [‡]
LLaVA-Med 1.5 7B	+0.02	+0.01	+0.01	-0.02	+0.00	-0.01
CheXagent-8B	-0.00	+0.01	-0.01	-0.03	-0.13	-0.03
Qwen2.5-VL-3B	+0.46	+0.42	+0.53	+0.68	+0.61	+0.39
Qwen2.5-VL-7B	+0.48	+0.75	+0.61	+0.87	+0.51	+0.81
Qwen3-VL-8B	+0.42	+0.51	+0.74	+0.79	+0.59	+0.43
Gemma 3 27B	+0.24	+0.36	+0.09	+0.28	+0.15	+0.75
Gemma 3 4B	+0.16	+0.54	+0.24	+0.34	+0.44	-0.27
LLaVA-1.5-7B	+0.06	+0.00	+0.01	+0.07	+0.01	+0.00
MedGemma 27B	+0.33	-0.04	+0.08	+0.18	+0.21	-0.04
LLaVA-1.5-13B	+0.05	+0.00 [‡]	+0.02	+0.08	+0.02	-0.03
Llama 3.2 Vision 11B	-0.03	-0.02	+0.02	-0.02	-0.04	+0.01
LLaVA-Rad-7B	-0.06	+0.01	-0.02	-0.06	-0.04	-0.00
Owned	21	19	19	20	18	16

Across the three tables, 38 of the 300 chest cells are owned and most chest contrasts are negative, against 113 of the 150 COCO cells for the same checkpoints under the same procedure. Figure 8 summarises the write matrices behind these contrasts (Appendix A.12), and Figure 9 shows the grades on single images (Appendix A.13).

A.6 What each analysis scores its writes against

Unless stated otherwise, an analysis uses the write matrix’s rule: the own write must exceed the 95th percentile of the 119-direction random family and the absolute sham, and the verdict comes from simultaneous max- T bounds on the contrasts with the competitors. 18 of the 20 graded analyses (5,844 cells) keep this reference and 19 keep this rule. The exceptions are the semantic endpoints, which take the maximum of 18 random directions of their own; the answer direction under held-out templates, which has the random 95th percentile in 80 of its 270 (cell, template) pairs and the sham alone in the rest; and the logit-scale grade, which decides by a percentile interval on the margin-scale ownership contrast. The probe refits, alternative constructions, displacement lifts, and expert-label refits (3,060 cells) take the sham of the seed-0 logistic normal.

A.7 Controls and robustness

Reading and support. Median probe selectivity over the chest cells is 0.14 (quartiles 0.08 and 0.22), and 141 of the 199 answerable chest cells are also readable. Effusion and Cardiomegaly are the most readable of the four concepts both chest sets share, with mean selectivity 0.18 and 0.19 over 50 chest blocks. Both grades need ten known positives and ten known negatives among the 400 calibration rows. CheXpert Atelectasis lacks them in each of the 25 CheXpert blocks (67 positives, 4 negatives), which makes 25 of the 300 chest cells ineligible, and COCO bicycle lacks them with 9 positives. Among eligible chest cells, 65% are readable and 72% answerable.

Dose and locus. The dose sweep ran on 25 NIH and 25 COCO blocks, and on COCO the ownership margins span 0.06–0.96 in the Qwen, Lingshu, and Gemma families. Writes at the connector locus are an order of magnitude smaller than at the consumed block: median $|W_{q,q}|$ is 0.008 against 0.032 on NIH and 0.008 against 0.127 on COCO.

Alternative directions. Besides the logistic normal, we write four constructions per concept (Table 6): the difference of class means, the Haufe pattern, the normal orthogonalised against the other five normals, and the normal refitted on features residualised against the other five labels. A classifier coefficient lifts by Equation 2. A difference of means or a pattern is instead a displacement $\mathbf{u}$ in the projected, train-scaled space, and its minimum-norm preimage gives the displacement lift

$$\hat{\mathbf{w}} \propto R(R^\top R)^{-1} \text{diag}(\mathbf{s}) \mathbf{u}, \quad \text{so that} \quad R^\top \hat{\mathbf{w}} \propto \text{diag}(\mathbf{s}) \mathbf{u}. \quad (5)$$

The two lifts differ because $R^\top R$ is not a multiple of the identity (Appendix F.3). The difference-of-means and pattern directions lie at a median absolute raw model-space cosine of 0.04 to 0.28 from the logistic normal.

Token weights and competitors. A token-weighted write replaces the uniform dose of Equation 1 with weights ω_t that average one over the n_{tok} consumed tokens,

$$\mathbf{h}'_{it} = \mathbf{h}_{it} + \alpha \omega_t \|\mathbf{h}_{it}\|_2 \hat{\mathbf{w}}_c, \quad \omega_t = \begin{cases} n_{\text{tok}} e^{z_t} / \sum_{t'} e^{z_{t'}} & \text{softmax,} \\ 4 \cdot \mathbf{1}[z_t \text{ in the top quarter}] & \text{top quarter,} \end{cases} \quad (6)$$

where z_t is the probe’s score of token t . The softmax variant keeps the uniform write’s L1 dose and the top-quarter variant doubles its Frobenius norm. The extended competitor family, scored across 9 blocks, adds a direction for every other label with at least 100 known positives and 100 known negatives (NIH: Consolidation, Edema, Infiltration; CheXpert: Enlarged Cardiomeastinum, Fracture, Lung Lesion, Lung Opacity, Pneumonia, Support Devices), fitted and written like the protocol normals; a cell is then owned only if its write also beats every added direction.

Table 6: **Alternative directions and competitor families.** Cells owned under the paper’s rule within each family, and the median O_q , per dataset. Each panel repeats the reference write on its own blocks.

Direction	Lift	NIH ChestX-ray14		CheXpert Plus		COCO	
		owned	med. O_q	owned	med. O_q	owned	med. O_q
<i>Direction construction (20 blocks per dataset)</i>							
logistic normal	coefficient	7/120	-0.03	19/120	-0.01	101/120	+0.18
difference of means	coefficient	16/120	-0.02	18/120	-0.01	62/120	+0.02
Haufe pattern	coefficient	17/120	-0.02	17/120	-0.01	53/120	+0.01
orthogonalised normal	coefficient	8/120	-0.02	14/120	-0.01	98/120	+0.13
residualised normal	coefficient	8/120	-0.02	13/120	-0.01	102/120	+0.20
owned under at least one		25/120	–	41/120	–	108/120	–
<i>Displacement lift (19 blocks per dataset)</i>							
logistic normal	coefficient	7/114	-0.03	19/114	-0.01	100/114	+0.19
difference of means	displacement	21/114	-0.03	19/114	-0.00	58/114	+0.02
Haufe pattern	displacement	17/114	-0.02	17/114	-0.00	49/114	+0.01
<i>Token weights of the logistic normal (4 NIH, 2 CheXpert, 2 COCO blocks)</i>							
uniform	coefficient	3/24	-0.08	5/12	+0.00	12/12	+0.60
softmax of the probe score	coefficient	0/24	-0.01	0/12	-0.00	12/12	+0.16
top quarter of tokens	coefficient	2/24	-0.12	3/12	-0.05	12/12	+0.57
<i>Competitor family of the logistic normal (5 NIH, 4 CheXpert blocks)</i>							
six clinical directions	coefficient	3/30	-0.07	8/24	-0.07	–	–
with every extra dataset label	coefficient	3/30	-0.07	5/24	-0.10	–	–

Geometry of the competitors. The strongest competitor coincides with the most similar direction or the most co-occurring label at about the chance rate of one cell in five, and the competitor’s advantage $W_{q,d} - W_{q,q}$ is uncorrelated with the raw cosine between the two directions and with the correlation ϕ between the two binary labels (Table 7).

NIH shows the deficit although its six directions are nearly orthogonal and its labels nearly uncorrelated, while on CheXpert directions and labels are correlated (Effusion–Consolidation cosine 0.46 averaged over blocks, $\phi = 0.74$) and the dominance is the same. At least two competitors exceed the own write in 54% of chest cells, so removing the strongest competitor leaves O_q negative in those cells. On COCO, competitors restricted to co-occurring objects (for example person and dog, $\phi = -0.06$) still leave $O_q > 0$ in 91–96% of each family’s cells, against 82–90% with all five. In the Effusion case, Nodule is the least similar direction to Effusion (raw cosine -0.06), yet it moves the Effusion answer by 0.262 against 0.190 for Effusion’s own direction.

Table 7: **Direction and label geometry against off-diagonal dominance.** One column per dataset, with the two chest sets pooled in the third. A “–” marks a row that is defined per dataset only.

	NIH ChestX-ray14	CheXpert Plus	both chest sets	COCO
<i>Geometry of the six directions and the six labels</i>				
mean off-diagonal cosine between the directions	0.019	0.215	–	0.043
mean off-diagonal $ \cos $ between the directions	0.093	0.218	–	0.058
mean $ \cos $ between random unit directions	0.024	0.024	–	0.024
mean off-diagonal ϕ between the labels	0.026	0.381	–	0.022
<i>Cells whose strongest competitor is (chance: one in five)</i>				
the most similar direction	31/150	33/150	64/300	31/150
the label with the largest ϕ	35/150	36/150	71/300	29/150
the most co-occurring label	24/150	32/150	56/300	22/150
<i>Spearman ρ over off-diagonal cells of $W_{q,d} - W_{q,q}$ with</i>				
the cosine between the two directions	0.085	0.048	0.037	0.042
the ϕ between the two labels	0.036	0.037	-0.001	0.007
<i>Sign of O_q</i>				
cells with $O_q > 0$	38/150	43/150	81/300	129/150
cells with $O_q > 0$ without the strongest competitor	62/150	75/150	137/300	136/150
cells with at least two competitors above the own write	88/150	75/150	163/300	14/150

Scale and saturation. Let O_q^m denote ownership recomputed from changes in the margin $\ell_{\text{yes}} - \ell_{\text{no}}$ instead of $P(\text{yes})$. The margin-scale grade reproduces the owned grade in 440 of 450 cells, every owned cell keeps $O_q^m > 0$, and the full margin-scale grade keeps 148 of the 151 owned cells. Rows with $P(\text{yes}) \notin [0.01, 0.99]$ are far more common in COCO blocks (median block share 0.71) than in NIH (0.16) and CheXpert (0.10) blocks. On unsaturated rows alone, all 38 owned chest cells keep $O_q > 0$ and 41 of the 252 others have $O_q > 0$; the full regrade keeps 36 of 38 owned chest cells and raises 0 of the 252 others to owned. The clean-No, clean-Yes, label-present, and label-absent strata give the same pattern.

Refits and numerics. Refitting the probes on resampled training patients and grading each refit against the seed-0 references changes O_q by a median of 0.02 (Table 8). Rescoring the first 200 test rows of each block in fp32 and at batch size one moves a written cell by at most 0.056 and changes 1 verdicts and 0 steering references among 108 grades. The bf16 batch effects declared at the gates (0.00–1.84 in candidate logits) leave the margin sign unchanged in 70 of 76 blocks, and the 6 others contain no owned cell. Re-scoring the clean baseline in a second module agrees bit for bit on every row in 44 of 74 blocks.

Table 8: **The full grade under refitted probes.** Cells per seed-0 grade (owned, stronger competitor, unresolved, fixed-family advantage without the reference) and how many keep it under both refits, one, or none.

Dataset	blocks cells		owned			stronger competitor			unresolved			adv. no ref.		seed 0
			<i>n</i>	both	≥ 1 none	<i>n</i>	both	≥ 1	<i>n</i>	both	≥ 1	<i>n</i>	both	
NIH ChestX-ray14	25	150	13	4	8	5	101	69	97	21	5	12	15	0 150/150
CheXpert Plus	25	150	25	6	17	8	95	63	89	21	3	10	9	3 150/150
COCO	25	150	113	99	110	3	17	13	17	10	4	7	10	2 149/150
<i>chest</i>	50	300	38	10	25	13	196	132	186	42	8	22	24	3 300/300
<i>all</i>	75	450	151	109	135	16	213	145	203	52	12	29	34	5 449/450

Radiologist labels. We re-grade the report-label directions on 200 films labelled by radiologist consensus, from patients in no campaign cohort. $|O_q(\text{valid}) - O_q(\text{test})|$ has a 90th percentile of 0.021, and on the 76 supported cells readability agrees in 73 of 76 and answerability in 60 of 76. On CheXpert, restricting to rows with a known label keeps the sign of O_q in 143 of 150 cells.

Refitting the six directions on the radiologist labels, cross-fitted over patients in 5 folds and written on the 600 test rows, gives directions at a median raw cosine of 0.12 (whitened 0.56) from the report-label ones, owned in 5 of 36 cells against 12. The refit reaches a held-out AUROC of 0.69 on the radiologist labels, against 0.79 for the report-label direction, and is fitted on 160 rows per fold against a median of 5332. Three further arms, scored on the same films with the same cross-fitting, separate the label source from the sample size: report-derived labels of those 200 films reach 0.67; report-derived labels of a training subsample with the refit’s 160 fitting rows, redrawn 20 times, reach 0.70; and every labelled training row reaches 0.79. With $\text{AUROC}_{L,n}$ the held-out AUROC against the radiologist labels of a direction fitted on n rows carrying labels L , $n = 160$ the refit’s sample and N every labelled training row,

$$\begin{aligned}\Delta_{\text{label}} &= \text{AUROC}_{\text{radiologist},n} - \text{AUROC}_{\text{report},n} = -0.01, \\ \Delta_{\text{size}} &= \text{AUROC}_{\text{report},N} - \text{AUROC}_{\text{report},n} = +0.09.\end{aligned}\tag{7}$$

The size-matched report refit also lies as far from the shipped direction as the radiologist refit (raw cosine 0.22 against 0.12).

A.8 The reader

Shared-tower pairs. Gemma 3 4B and 12B write bit-identical vectors, and the other Gemma 3 and MedGemma pairs agree to bf16 precision (Table 9). On NIH Effusion, the identical write

gives Gemma 3 12B and 27B an O_q difference of +0.79 [+0.76, +0.82]. In 12B, competing directions win the chest answers outright ($O_{\text{Effusion}} = -0.83$ on both chest sets, $O_{\text{Mass}} = -0.65$ on NIH, $O_{\text{Pneumothorax}} = -0.71$ on CheXpert); in 27B, no direction moves the same answers ($-0.04, -0.08, -0.01$). Gemma 3 4B answers yes on 94–97% of chest rows, so many of its cells sit at ceiling; on the margin scale its O_q^m runs from -1.1 to -19.8 with every interval below zero, so every own write lowers its answer’s margin more than any competitor raises it. MedGemma 4B and 27B differ in the same way (CheXpert Pneumothorax -0.60 against -0.06 , Edema $+0.10$ against -0.28).

Table 9: **Same write, different reader: paired differences.** $\Delta O_q = O_q(\text{larger}) - O_q(\text{smaller})$ per shared-tower pair on the same 600 test rows, with paired patient-bootstrap 95% intervals; the last two columns divide $|\Delta O_q|$ by the two blocks’ mean refit standard deviation.

Pair	Dataset	relation	ΔO_q CI	$ \Delta O_q $		ratio to refit SD	
			$\neq 0$	median	max	median	max
Gemma 3 4B → Gemma 3 12B	NIH ChestX-ray14	bit-identical	5/6	0.33	0.83	2.7	6.9
Gemma 3 4B → Gemma 3 12B	CheXpert Plus	bit-identical	6/6	0.29	0.83	2.1	5.9
Gemma 3 4B → Gemma 3 12B	COCO	bit-identical	5/6	0.23	0.87	1.6	6.0
Gemma 3 12B → Gemma 3 27B	NIH ChestX-ray14	bf16-equal	6/6	0.37	0.79	3.7	8.0
Gemma 3 12B → Gemma 3 27B	CheXpert Plus	bf16-equal	6/6	0.35	0.81	1.4	3.3
Gemma 3 12B → Gemma 3 27B	COCO	bf16-equal	5/6	0.15	0.40	1.7	4.3
Gemma 3 4B → Gemma 3 27B	NIH ChestX-ray14	bf16-equal	5/6	0.07	0.24	1.8	6.0
Gemma 3 4B → Gemma 3 27B	CheXpert Plus	bf16-equal	6/6	0.17	0.65	4.6	17.3
Gemma 3 4B → Gemma 3 27B	COCO	bf16-equal	6/6	0.16	1.02	1.4	8.8
MedGemma 4B → MedGemma 27B	NIH ChestX-ray14	bf16-equal	5/6	0.09	0.30	1.5	5.0
MedGemma 4B → MedGemma 27B	CheXpert Plus	bf16-equal	5/6	0.19	0.54	1.8	5.0
MedGemma 4B → MedGemma 27B	COCO	bf16-equal	3/6	0.05	0.35	0.6	4.6

Size and family. Chest ownership concentrates in a few readers and varies by concept and dataset (Figure 7; Table 2). Qwen2.5-VL owns no concept at 3B or 7B, Cardiomegaly at 32B, and Atelectasis ($O_{\text{Atelectasis}} = 0.54$), Cardiomegaly, and Mass at 72B; Effusion has negative ownership at every size. Lingshu, the medical Qwen2.5-VL derivative, owns three CheXpert concepts at 7B and five at 32B, the most in the grid. LLaVA-1.5 reads the chest concepts but answers none above chance and owns none.

Chest-trained towers. The towers of CheXagent-2-3B, CheXagent-8B, and LLaVA-Rad-7B are trained on chest radiographs; HuatuoGPT-Vision-7B is tuned on medical data over a stock CLIP tower. The chest-trained towers reach a median probe AUROC of 0.855 across 18 NIH cells, against 0.773 for HuatuoGPT-Vision and 0.774 for the remaining 132, and they read 29 and answer 32 of their 36 chest cells. On NIH, 2 of 18 cells are owned for chest-trained towers, 4 of 36 for medically tuned checkpoints, and 7 of 96 for general checkpoints. The COCO control rests on checkpoints that answer natural-image questions: of the COCO cells, 80 of 96 are answerable and 84 owned for the general group (median clean-answer AUROC 0.987), 30 of 36 and 27 for the medically tuned group (0.962), and 6 of 18 and 2 for the chest-trained towers (0.538).

Crossed tower and reader. We load the vision tower of Gemma 3 4B (G) behind the connector and language model of MedGemma 4B (M), and the reverse. All 437 weight tensors of the loaded tower equal the donor’s bit for bit, and none of the 451 tensors outside it changes. Directions travel with their tower, so changing the tower changes the representation and its six directions together, while changing the reader changes only the connector and the language model. All 4 combinations are graded on the same 200 test rows of the block, at the primary dose and template, against the 119-direction random family and the sham of the tower in use. The two native combinations reproduce the checkpoints’ own grades on this smaller cohort: across the 36 native cells, the ownership contrast moves by at most 0.060, and 3 of the 13 owned cells become unresolved. Writing $Y_q(g/h)$ for a

quantity of question q with tower g and reader h , the mean reader and tower effects are

$$\delta_{\text{reader}} = \frac{1}{12} \sum_g \sum_q |Y_q(g/G) - Y_q(g/M)|, \quad \delta_{\text{tower}} = \frac{1}{12} \sum_h \sum_q |Y_q(G/h) - Y_q(M/h)|, \quad (8)$$

for $Y_q \in \{W_{q,q}, O_q, M_q\}$, where $M_q = W_{q,q} - \max(\max_{d \neq q} W_{q,d}, p_{95}, |W_q^{\text{sham}}|)$ is the own write's margin over the whole reference family, with p_{95} the random 95th percentile and W_q^{sham} the sham write.

Under the four combinations, 0 of 48 chest cells are owned, against 14 of 24 COCO cells: on COCO, 5 of the six objects are owned under each native combination, 4 with the Gemma tower behind the MedGemma reader, and 0 with the MedGemma tower behind the Gemma 3 reader. On NIH, the reader effect is 0.186 [0.170, 0.207] on $W_{q,q}$ and 0.184 [0.169, 0.203] on O_q , against 0.162 and 0.163 for the tower; on CheXpert it is 0.254 and 0.259, against 0.226 and 0.260; and on the family margin M_q the two chest sets give 0.356 and 0.365 for the reader against 0.316 and 0.444 for the tower. On COCO the tower effect is the larger on all three quantities (0.248, 0.416, and 0.353 against 0.137, 0.327, and 0.288).

The hybrid combinations answer worse than the native ones. With no write applied, all 6 of the 6 hybrid arms have a lower median clean-answer AUROC than the native arm of their block, by 0.081 to 0.442: it falls from 0.930 to 0.488 when the MedGemma tower serves the Gemma 3 reader on COCO, and from 0.840 to 0.549 and from 0.889 to 0.636 when the Gemma 3 tower serves the MedGemma reader on NIH and CheXpert.

Identical-tensor replay. Checkpoints of one family share a vision tower, but two readers' inputs agree only up to bf16 rounding. The replay module stores the consumed block's token tensors once from the group's source checkpoint and feeds them to every reader of the group in place of that reader's own tower output, so that the readers consume the same bits. Before the replacement it measures the drift, the largest absolute difference between the reader's own tower output and the stored tensor on the same rows. In 7 of the 8 replays that output already matches the stored tensor bit for bit; the exception is LLaVA-Med, whose fine-tuned tower departs by up to 112.16 against a mean absolute tensor entry of 0.62–29.12.

A reader that replays its own block's tensors tests the replacement path itself: across the 3 such replays no verdict and no ownership grade changes, 2 reproduce the block's grade exactly, and the third differs by at most 6.2×10^{-3} in a write magnitude. Across all 8 replays, the change is distinguishable from zero in 5 of 48 concept cells, the largest change in a write magnitude is 1.5×10^{-2} , and 0 verdicts and 0 ownership grades change. For the 7 reader pairs, the mean absolute difference between the two readers' own-question writes is 0.016–0.348 when each reads its own tower, and feeding both the stored tensors changes that difference by at most 0.0052, or 1.9% of its size.

A.9 Non-clinical attributes

The three attribute labels come from each dataset's metadata: anteroposterior or portable view against posteroanterior, recorded sex, and age at least 60. Their directions are fitted, lifted, and written with the six clinical directions as one nine-direction family, against the same random family and sham in all 38 blocks. All 114 of the 114 attribute cells are readable, with median probe AUROC 0.896–0.999 against the label (Table 10). Each attribute question is scored in 3 phrasings on the calibration rows; the written phrasing reaches a median clean-answer AUROC of 0.679, against 0.716 for the best phrasing of the cell, and 100 of the 114 attribute cells are answerable. Attributes are owned no more often than findings under three comparisons: over every cell (12 of 114 against 23 of 228), over answerable cells (11 of 100 against 22 of 141), and over 188 pairs matched within 0.05 clean-answer AUROC in the same block (14 against 17). Over the 24 attribute–finding pairs, the largest absolute raw cosine between an attribute direction and a clinical normal is 0.11, for AP (portable) projection with Edema.

Table 10: **Non-clinical attributes of the same radiographs.** Per attribute and dataset, medians over blocks of the attribute probe’s AUROC and selectivity, the clean-answer AUROC of the attribute question, and the absolute raw cosine to the clinical normals, with the owned attribute cells.

Attribute	Dataset	probe AUROC	selectivity	answer AUROC	med. cos	owned
AP (portable) projection	NIH ChestX-ray14	0.999	0.32	0.76	0.03	1/19
AP (portable) projection	CheXpert Plus	0.998	0.31	0.73	0.05	0/19
AP (portable) projection	<i>chest</i>	0.999	0.31	0.74	0.04	1/38
female sex	NIH ChestX-ray14	0.984	0.30	0.73	0.05	3/19
female sex	CheXpert Plus	0.957	0.27	0.65	0.06	1/19
female sex	<i>chest</i>	0.971	0.28	0.71	0.05	4/38
age ≥ 60	NIH ChestX-ray14	0.896	0.21	0.62	0.05	3/19
age ≥ 60	CheXpert Plus	0.898	0.21	0.64	0.06	4/19
age ≥ 60	<i>chest</i>	0.896	0.21	0.63	0.06	7/38

A.10 The answer direction

Fitting and writing. For question q , let $m_{iq} = \ell_{\text{yes}} - \ell_{\text{no}}$ be the clean answer margin of training image i on the block’s primary template, and $\mathbf{x}_i$ its projected, train-scaled features. The answer direction a_q lifts, by Equation 2, the ridge coefficients

$$\gamma_q = \arg \min_{\gamma_0, \gamma} \sum_i (m_{iq} - \gamma_0 - \mathbf{x}_i^\top \gamma)^2 + \lambda \|\gamma\|_2^2, \quad (9)$$

fitted on the first 3,000 training rows in the manifest’s label-blind order, with $\lambda \in \{0.1, 1, 10, 100\}$ chosen by five-fold cross-validated R^2 over patients or images. We write a_q at the same dose against the write matrix’s random family and a sham of the a_q under test, with the other five a_d as competitors. Under the five held-out templates, a_q is written unchanged against each template’s own clean baseline and sham.

Where it is owned. The answer direction beats the strongest logistic competitor in 30 chest cells (Table 11). Over all held-out (cell, template) pairs it is owned in 87 of 180 chest and 89 of 90 COCO pairs, and it stays owned in 89 of 90 COCO pairs owned under the primary template. In Qwen2.5-VL-7B on NIH, the Effusion answer direction reaches $O_q = +0.29$ with an own write of 0.81.

Table 11: **Ownership under the answer direction.** Per block, on the primary template: concepts owned by the label direction $\hat{w}_q$ and by the answer direction a_q , split into kept (both), rescued (a_q only), and lost ($\hat{w}_q$ only); cells where the write of a_q beats the strongest logistic competitor; and medians over the six questions of the own write of a_q , $W_{q,q}^a$, and of the ridge fit’s R^2 .

Checkpoint	Dataset	owned concepts					$a_q > \text{comp.}$	median	
		$\hat{w}_q$	a_q	kept	rescued	lost		$W_{q,q}^a$	R^2
Qwen2.5-VL-7B	NIH ChestX-ray14	0/6	3/6	0	3	0	6/6	0.61	0.54
Qwen2.5-VL-7B	CheXpert Plus	0/6	6/6	0	6	0	6/6	0.55	0.48
Qwen2.5-VL-7B	COCO	6/6	6/6	6	0	0	6/6	0.78	0.58
Gemma 3 12B	NIH ChestX-ray14	1/6	2/6	1	1	0	5/6	0.47	0.17
Gemma 3 12B	CheXpert Plus	1/6	0/6	0	0	1	1/6	0.37	0.19
Gemma 3 12B	COCO	6/6	6/6	6	0	0	6/6	0.61	0.37
Lingshu 32B	NIH ChestX-ray14	2/6	3/6	2	1	0	6/6	0.24	0.77
Lingshu 32B	CheXpert Plus	5/6	5/6	5	0	0	6/6	0.22	0.80
Lingshu 32B	COCO	6/6	6/6	6	0	0	6/6	0.36	0.55
NIH ChestX-ray14	3 blocks	3/18	8/18	3	5	0	17/18	0.47	0.54
CheXpert Plus	3 blocks	6/18	11/18	5	6	1	13/18	0.35	0.48
<i>chest</i>	6 blocks	9/36	19/36	8	11	1	30/36	0.41	0.50
COCO	3 blocks	18/18	18/18	18	0	0	18/18	0.61	0.54

What it reads. In Qwen2.5-VL-7B on NIH (Table 12), the answer direction’s AUROC is 0.60 against the label and 0.93 against the clean answer, against 0.73 and 0.66 for the probe. Cosines depend on the inner product chosen for the representation (Park et al., 2024), so besides the raw cosine we report, for two coefficient vectors β, β' in the projected space,

$$\cos_{\Sigma}(\beta, \beta') = \frac{\beta^{\top} \Sigma \beta'}{\sqrt{\beta^{\top} \Sigma \beta \beta'^{\top} \Sigma \beta'}}, \quad (10)$$

with Σ the covariance of the training features: the correlation over training images between the two vectors’ scores. Between answer and probe coefficients, the median raw cosine is 0.06–0.07 on the chest blocks of Qwen2.5-VL-7B, 0.30–0.33 on those of Lingshu 32B, and 0.54–0.64 on COCO; the median whitened cosine is 0.36–0.43 on the chest sets and 0.81 on COCO, and the two metrics differ most for Lingshu 32B on CheXpert (0.33 raw, 0.87 whitened).

The column selectivity

$$\kappa_d = \frac{W_{d,d}}{\sum_q |W_{q,d}|} \quad (11)$$

equals one when a write moves only its own answer. Over the 36 chest cells its median is 0.24 for answer directions and 0.19 for label directions, against 0.78 and 0.73 over 18 COCO cells; over all cells it is 0.12 on NIH, 0.12 on CheXpert, and 0.57 on COCO. On the chest sets, both directions move the other five answers along with their own.

Table 12: **The answer direction against the label direction.** Per block, medians over the six questions of the AUROCs of a_q and $\hat{w}_q$ against the label and against the clean answer, the Pearson correlation, and the raw and whitened cosines between a_q and $\hat{w}_q$.

Checkpoint	Dataset	AUROC				Pearson	cosine	
		label		clean answer			raw	Σ
		a_q	$\hat{w}_q$	a_q	$\hat{w}_q$			
Qwen2.5-VL-7B	NIH ChestX-ray14	0.60	0.73	0.93	0.66	0.36	0.07	0.35
Qwen2.5-VL-7B	CheXpert Plus	0.67	0.86	0.85	0.66	0.34	0.06	0.36
Qwen2.5-VL-7B	COCO	0.97	0.97	0.98	0.98	0.84	0.64	0.84
Lingshu 32B	NIH ChestX-ray14	0.75	0.76	0.95	0.79	0.62	0.30	0.60
Lingshu 32B	CheXpert Plus	0.85	0.86	0.95	0.89	0.88	0.33	0.87
Lingshu 32B	COCO	0.94	0.97	0.93	0.92	0.75	0.54	0.73
Gemma 3 12B	NIH ChestX-ray14	0.59	0.65	0.72	0.61	0.31	0.18	0.33
Gemma 3 12B	CheXpert Plus	0.61	0.78	0.73	0.56	0.13	0.17	0.10
Gemma 3 12B	COCO	0.93	0.96	0.80	0.76	0.81	0.59	0.79

Semantic endpoints. Four endpoints keep the write and change what is read after it: the negated question, whose effect must take the opposite sign; the forced choice between the finding and its strongest competitor, asked in both orders so that a position preference shows as a gap; and the log-probability of the finding word in a report continuation. Each carries its own clean baseline, sham, and 18 random directions. With E_q the signed endpoint effect of the own write, a cell meets the reference when

$$E_q > \max\left(0, \max_k E_k^{\text{rand}}, |E_q^{\text{sham}}|\right), \quad (12)$$

the inner maximum running over the 18 random directions, and it is owned when it also holds the 6×5 max- T advantage.

Across 3 blocks, the negated question moves opposite to the affirmative one in 16 of 18 concept-block pairs, the forced choice is owned in both orders in 3 of 18 pairs (mean order gap 0.026–0.111), and the report continuation is owned in 0 of 18 cells (Table 13). Adding the ownership contrast to a regression of the endpoint effect on write magnitude, probe selectivity, and clean-answer AUROC changes the held-out R^2 by +0.687 when one concept is left out and +0.070 when one of 3 blocks is

left out; permuting ownership within blocks 2,000 times gives $p = 0.26$ and $p = 0.26$. The predictors vary by concept, so each block contributes an effective sample of six.

Table 13: **Four endpoints no direction was fitted against.** Per block and endpoint, the concepts of six whose own write is owned and whose own write meets the steering reference, for the label direction and the answer direction.

Checkpoint	Dataset	Endpoint	label direction		answer direction	
			owned	reference	owned	reference
Lingshu 32B	CheXpert Plus	negated question	2/6	2/6	3/6	4/6
Lingshu 32B	CheXpert Plus	forced choice, finding first	3/6	3/6	4/6	4/6
Lingshu 32B	CheXpert Plus	forced choice, competitor first	4/6	4/6	5/6	5/6
Lingshu 32B	CheXpert Plus	report continuation	0/6	0/6	0/6	0/6
Qwen2.5-VL-7B	NIH ChestX-ray 14	negated question	0/6	1/6	3/6	6/6
Qwen2.5-VL-7B	NIH ChestX-ray 14	forced choice, finding first	0/6	0/6	3/6	3/6
Qwen2.5-VL-7B	NIH ChestX-ray 14	forced choice, competitor first	0/6	0/6	2/6	2/6
Qwen2.5-VL-7B	NIH ChestX-ray 14	report continuation	0/6	0/6	0/6	1/6
Qwen2.5-VL-7B	CheXpert Plus	negated question	1/6	1/6	5/6	6/6
Qwen2.5-VL-7B	CheXpert Plus	forced choice, finding first	0/6	0/6	3/6	4/6
Qwen2.5-VL-7B	CheXpert Plus	forced choice, competitor first	0/6	0/6	2/6	2/6
Qwen2.5-VL-7B	CheXpert Plus	report continuation	0/6	0/6	0/6	0/6
<i>all</i>	<i>3 blocks</i>	<i>4 endpoints</i>	10/72	11/72	30/72	37/72

A.11 Difficulty

Fine-grained objects. Six further COCO categories test whether ownership on COCO depends on large, easy objects. A rule stated in full in Appendix F.6 picks them from the label counts and mean instance areas of the fixed cohorts alone, and the protocol file registers the rule and the two matching windows before the module’s first outcome. The categories are fitted, written, and graded like the six easy objects, as their own family in the same blocks and rows, with their clean-answer AUROC measured on the same calibration rows. Within a checkpoint the owned counts of fine-grained and easy objects differ by -2 to 0 of six, and median probe selectivity is 0.058 for the fine-grained objects against 0.065 for the easy ones.

Difficulty matching. Each chest cell c is paired with the natural-image cells $\mathcal{P}(c)$ whose matching variables lie within 0.05 of its own. With $o \in \{0, 1\}$ a cell’s owned indicator, C the chest cells with at least one partner, and $\mathcal{P} = \bigcup_{c \in C} \mathcal{P}(c)$,

$$\Delta_{\text{pool}} = \frac{1}{|C|} \sum_{c \in C} o_c - \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} o_p, \quad \Delta_{\text{match}} = \frac{1}{|C|} \sum_{c \in C} \left(o_c - \frac{1}{|\mathcal{P}(c)|} \sum_{p \in \mathcal{P}(c)} o_p \right); \quad (13)$$

Δ_{pool} compares the two matched pools, while Δ_{match} weights every chest cell equally however many partners it has; neither can be obtained from the other. Chest cells without a partner leave the matching, which the stratified comparison below avoids. Intervals come from 5,000 draws of a cluster bootstrap over models (Table 14). Matched on clean-answer AUROC, Δ_{match} shrinks to -0.066 $[-0.304, -0.004]$, and matched on both variables, which keeps 212 of the 300 chest cells, its interval $[-0.198, +0.076]$ includes zero. The stratified comparison of Section 5.3 keeps every cell instead, binning both pools by clean-answer AUROC at the edges 0.50 , 0.70 , 0.80 and 0.90 ; the 37 chest cells of the top bin are all CheXpert Plus cells, since no NIH ChestX-ray 14 cell reaches 0.90 .

A.12 Write structure across the grid

Figure 8 summarises the 6×6 write matrices across all completed checkpoints as the cell-wise median matrix for each dataset. Figure 3 reads the same matrices as flow, with band width proportional to

Table 14: **Difficulty matching.** Per matching variable: matched chest cells, natural-image partner cells, and the pooled and matched differences of Equation 13 with 95% cluster-bootstrap intervals over models.

Matching	chest cells	partners	Δ_{pool} [95% interval]	Δ_{match} [95% interval]
probe selectivity alone	300/300	186/186	-0.642 [-0.763, -0.512]	-0.539 [-0.678, -0.359]
clean-answer AUROC alone	300/300	186/186	-0.642 [-0.749, -0.505]	-0.066 [-0.304, -0.004]
both variables	212/300	101/186	-0.487 [-0.653, -0.201]	-0.024 [-0.198, 0.076]
both, easy partners only	211/300	79/150	-0.432 [-0.625, -0.150]	-0.008 [-0.198, 0.085]

$F_{q,d}$, the mean over checkpoints of $\max(W_{q,d}, 0)$. The own-question share of that flow is

$$\frac{\sum_d F_{d,d}}{\sum_{q,d} F_{q,d}} = 24\% \text{ (NIH)}, \quad 22\% \text{ (CheXpert)}, \quad 79\% \text{ (COCO)}. \quad (14)$$

It pools positive changes over checkpoints, whereas the column selectivity κ_d of Equation 11 keeps the sign of the own effect, counts negative changes, and is summarised per cell. The largest sinks on the chest sets, questions that every direction moves, are Cardiomegaly on NIH and Edema and Effusion on CheXpert.

A.13 Per-image examples

Figure 9 shows grades for individual test images, with $P(\text{yes})$ in parentheses on every answer line. The first four chest examples are deficit cells of Qwen2.5-VL-7B, the next two are owned cells (Lingshu 32B Edema, Qwen2.5-VL-72B Atelectasis), and the natural-image examples are owned cells of Qwen2.5-VL-7B, Lingshu 32B, and MedGemma 4B. For a chest example, we pool test rows where the finding is labelled present, the probe reads it (probability above 0.5), and the clean answer is No. For a natural-image example, we pool rows where the probe reads the label correctly and the clean answer is No. We split each pool by whether the concept’s own write or the strongest competing write raises the answer more. Among rows where either write raises the answer by at least 0.2, we show the row with the gap closest to the median. Each example’s last line reports how many pool rows share its ordering, giving the example its own base rate.

B Pilot study: registered protocol

The two-model study that motivated the evaluation registered the procedures below; Appendix C reports its results. It uses the construction of Section 3, with the differences stated here.

Data, models, and reading. We use NIH ChestX-ray14 (Wang et al., 2017), whose findings are mined from radiology reports. A deterministic hash of patient identity assigns rows to disjoint train, validation, and test partitions: 18,212 training images and 5,343 test images from 1,659 test patients. The three cells pair LLaVA-1.5-7B (Liu et al., 2024) with Effusion and Edema, and Qwen2.5-VL-7B (Bai et al., 2025b) with Effusion, at the final visual block consumed by the connector or merger; hook tests confirm that this block lies on the downstream path, that $\alpha = 0$ is a bitwise no-op, and that nonzero doses change downstream logits. Projection, scaling, and probe follow Section 3.2 with seed 0. Selectivity is Equation 3 on test rows, with 20 control seeds mapping the 39 recurring input types (AP view, sex, and age decade) to random labels and 2,000 patient-cluster bootstrap draws. Selectivity compares the concept with nuisance-type label maps of the same capacity; the write tests whether the model uses it. All uncertainty clusters by patient; prompts appear in Appendix D.

Write and doses. The write is Equation 1 along the lifted normal of Equation 2, with every entry of s floored at 10^{-8} , and the endpoint is the paired mean change in $P(\text{yes})$ from the common unsteered baseline (Section 3.3). Original cells use the doses $\{-1, -0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5, 1\}$ on 200 held-out test images. At the six controlled doses $\mathcal{A} = \{\pm 1, \pm 0.5, \pm 0.25\}$, 20 isotropic random unit directions, one coordinate-permutation sham, and the fixed alternative clinical normals receive the same signed dose on the same rows. Dose monotonicity over $[-0.5, 0.5]$ is descriptive.

Table 15: **Pilot replication on 600 new patients.** Qwen2.5-VL-7B on NIH ChestX-ray14. Per concept: the concept write $W_{q,q}$, the random 95th percentile (p95), the absolute sham, and its rank among the 120 compared effects; blue shades a negative ownership contrast.

Concept	S	AUROC	$W_{q,q}$	p95	sham	rank	O_q [95% CI]	Competitor	Verdict
Effusion	0.102	0.593	+0.190	0.137	0.017	2/120	-0.071 [-0.080, -0.062]	Nodule	competitor
Atelectasis	0.062	0.566	+0.101	0.297	0.096	55/120	-0.226 [-0.233, -0.219]	Nodule	competitor
Pneumothorax	0.152	0.508	+0.089	0.320	0.113	52/120	-0.102 [-0.108, -0.096]	Effusion	competitor
Cardiomegaly	0.205	0.742	+0.085	0.311	0.158	33/120	-0.219 [-0.227, -0.211]	Atelectasis	competitor
Mass	0.021	0.678	+0.359	0.372	0.092	10/120	-0.169 [-0.178, -0.160]	Effusion	competitor
Nodule	-0.041	0.642	+0.118	0.158	0.055	15/120	-0.072 [-0.081, -0.063]	Effusion	competitor

Original-cell decision rule. With $\Delta_c(\alpha)$ the target direction’s paired mean change at dose α , the primary effect and the dose that attains it are

$$\Delta_c^* = \max_{\alpha \in \mathcal{A}} \text{sign}(\alpha) \Delta_c(\alpha), \quad \alpha^* = \arg \max_{\alpha \in \mathcal{A}} \text{sign}(\alpha) \Delta_c(\alpha). \quad (15)$$

A cell clears the generic gate when Δ_c^* exceeds both the 95th percentile of the 20 random directions’ changes at α^* , each multiplied by $\text{sign}(\alpha^*)$, and the largest absolute sham effect over $\mathcal{A}$. The random directions are evaluated at α^* alone, so the gate is a registered reference without cross-dose correction.

Direction-specificity supplement. The Nodule response motivated a separately registered supplement at the locked Qwen dose $\alpha = +0.25$, on one row per patient, chosen by label-blind hash, for 400 ChestX-ray14 test patients absent from the original cohort. It estimates the Effusion row of the write matrix (Section 3.3) against the five other pre-existing Qwen normals (Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule), fitted like Effusion, together with 20 random directions and the sham: 28 configurations and 11,200 per-image outcomes. Direction specificity requires an Effusion effect above the random 95th percentile and the absolute sham, and a one-sided 95% lower bound of O_{Effusion} (Equation 4) above zero, with the maximum recomputed in each of 5,000 patient-bootstrap draws. Adding competitors can only lower O_q , so a positive advantage holds for the family tested, while a negative one needs no exhaustive family.

Causal-ownership matrix. A prospective grid crosses six Qwen directions (Effusion, Atelectasis, Pneumothorax, Cardiomegaly, Mass, and Nodule) with their six questions on 400 new patients at $\alpha = +0.25$, with 119 random directions and a sham. Question q is owned when $W_{q,q}$ exceeds every other clinical direction, every random direction, and the absolute sham, with the 30 clinical contrasts bounded simultaneously by Equation 16. A separate family of 12 statistics tests shared aliasing: for each direction, its signed mean effect on its five off-diagonal questions and its mean excess over the matching diagonals there.

Answer-encoding crossover and Mass confirmation. A registered crossover on 200 patients of the causal-ownership cohort, with the other 200 as a patient-disjoint calibration set, crosses the original yes/no prompt with A-present and B-present prompts that exchange the finding-to-letter mapping, at $\alpha = \pm 0.25$ with six clinical directions, 20 random directions, and a sham; a question enters only if both coded mappings have clean-answer AUROC lower bounds above 0.5 with at least ten patients per class. An independent confirmation then tests Mass on 200 patients absent from every earlier Qwen cohort, crossing two wordings (*show*, *is*) with explicit yes/no, A-present, and B-present instructions at $\alpha = +0.25$, with 119 random directions, a sham, and 20 clinical comparisons under simultaneous 95% lower bounds. Confirmation requires positive bounds and Mass effects above every random effect and the absolute sham, across all four coded cells or both original-wording cells.

C Pilot study: results

C.1 A non-generic write effect reproduces

The first finding starts with a successful intervention: Qwen Effusion reproduces beyond random and sham controls before clinical alternatives change its attribution.

Controlled decodability is evaluated at the final visual block each connector consumes: the penultimate CLIP encoder layer for LLaVA and the last of Qwen’s 32 visual blocks. Intervals are 95% patient-cluster bootstrap intervals over 2,000 draws; the control AUROC is the mean over the 20 recurring-type random-label maps, with its 5th–95th percentile spread in parentheses.

Qwen Effusion first satisfies the reader test. Its probe reaches AUROC 0.7742 [0.7488, 0.7993] against a control AUROC of 0.6576 (0.5738–0.7476), a controlled selectivity of 0.1166 (95% CI [0.0897, 0.1428]). LLaVA-1.5-7B reads both of its concepts at its consumed block: Effusion at AUROC 0.7788 [0.7538, 0.8031] with selectivity 0.1139 [0.0875, 0.1404], and Edema at AUROC 0.8009 [0.7648, 0.8329] with selectivity 0.1360 [0.0994, 0.1693], against a control AUROC of 0.6649 (0.5750–0.7450). Original-cohort intervention outcomes maximize the concept-consistent effect over six fixed nonzero doses. Qwen makes a strong write-direction case in the original 200-image cohort: the maximum concept-consistent change is 0.2343 at $\alpha = +0.25$, above the random-direction 95th percentile of 0.0963 and the maximum absolute sham effect of 0.0852.

The response at the locked dose $\alpha = +0.25$ reproduces on the patient-disjoint 400-patient cohort. Effusion changes mean $P(\text{yes})$ by 0.1945, again above the random-direction 95th percentile of 0.0878 and absolute sham effect of 0.0157. None of these patients appears in the original intervention cohort, and the dose was not reselected, supporting a large and reproducible non-generic effect within ChestX-ray14.

C.2 Semantic competition changes attribution, not efficacy

The fixed clinical family provides a harder identification test than non-semantic controls. Nodule changes the Effusion answer by 0.2519 and realizes the point-estimate maximum among the five alternatives. The Effusion ownership contrast O_{Effusion} is -0.0573 . The registered fixed-family endpoint recomputes the maximum inside each of 5,000 patient-bootstrap draws, giving a two-sided 95% CI of $[-0.0694, -0.0450]$ and a one-sided 95% lower bound of -0.0678 . The entire interval is below zero. Descriptively, all five alternative clinical normals move the Effusion endpoint in the same positive direction; Effusion exceeds four by 0.1172–0.1778, while Nodule alone has a larger point estimate. This shared-sign pattern motivates an answer-sensitive-alias hypothesis but cannot distinguish it from question-specific ownership.

Qwen Effusion therefore remains a reproducible direction that changes the answer beyond random and sham perturbations, while its specific semantic attribution fails. The familywise endpoint establishes that at least one fixed clinical alternative is stronger; Nodule is the point-estimate maximizer, not a separately familywise-identified semantic owner.

C.3 Wording changes clinical competition for Mass

In the registered crossover, the four questions that pass capability calibration (Atelectasis, Pneumothorax, Mass, and Nodule) respond more in the component that reverses the raw A-minus-B sign when the finding-to-letter mapping is exchanged, but so do all 20 random directions and the sham, so aggregate mapping response does not separate clinical normals from generic perturbations. Mass is the exception: its clinical margin, the Mass effect minus the largest of the five other clinical effects in finding-present logit units, changes from -0.7181 under the original yes/no prompt to 0.2825 and 0.2900 under A-present and B-present (Table 16, top). The independent confirmation on 200 new patients reproduces the clinical advantage under both *show* coded prompts, with simultaneous lower bounds of 0.1675 and 0.1462, while both *is* margins are negative and the Mass effects stay below the maximum of the 119 random directions (rank 3 of 120), so neither registered specificity conjunction is met (Table 16, bottom). Mass steering increases Brier error in every cell, and every AUROC-change interval spans zero.

Table 16: **Mass clinical competition across prompts.** Clinical margin per prompt: the Mass effect minus the largest of the five other clinical effects, in finding-present logit units. Top, the discovery crossover; bottom, the registered confirmation on new patients.

Prompt	Clinical margin	95% CI	Mass effect	Random max	Sham
Standard yes/no	-0.7181	[-0.7956, -0.6412]	2.8625	3.0412	0.5869
A-present	0.2825	[0.2269, 0.3406]	1.6125	1.1650	0.7550
B-present	0.2900	[0.2300, 0.3506]	1.7469	1.2581	1.1475
Condition	Clinical margin	Minimum LCB	Mass effect	Random max	
<i>show</i> /A-present	0.2525	0.1675	1.7019	2.2100	
<i>show</i> /B-present	0.2338	0.1462	1.8188	2.2169	
<i>is</i> /A-present	-0.1087	-0.1751	1.9225	2.4750	
<i>is</i> /B-present	-0.2006	-0.2686	2.0019	2.4944	

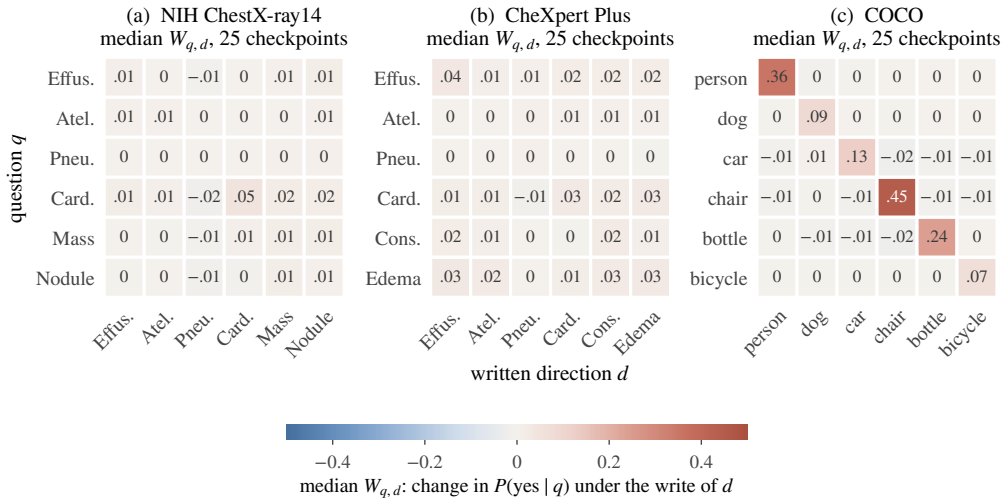


Figure 8: **Write structure across the grid.** Cell-wise median of the write matrix $W_{q,d}$ over completed checkpoints on NIH ChestX-ray14, CheXpert Plus, and COCO. Figure 3 reads the same matrices as flow.

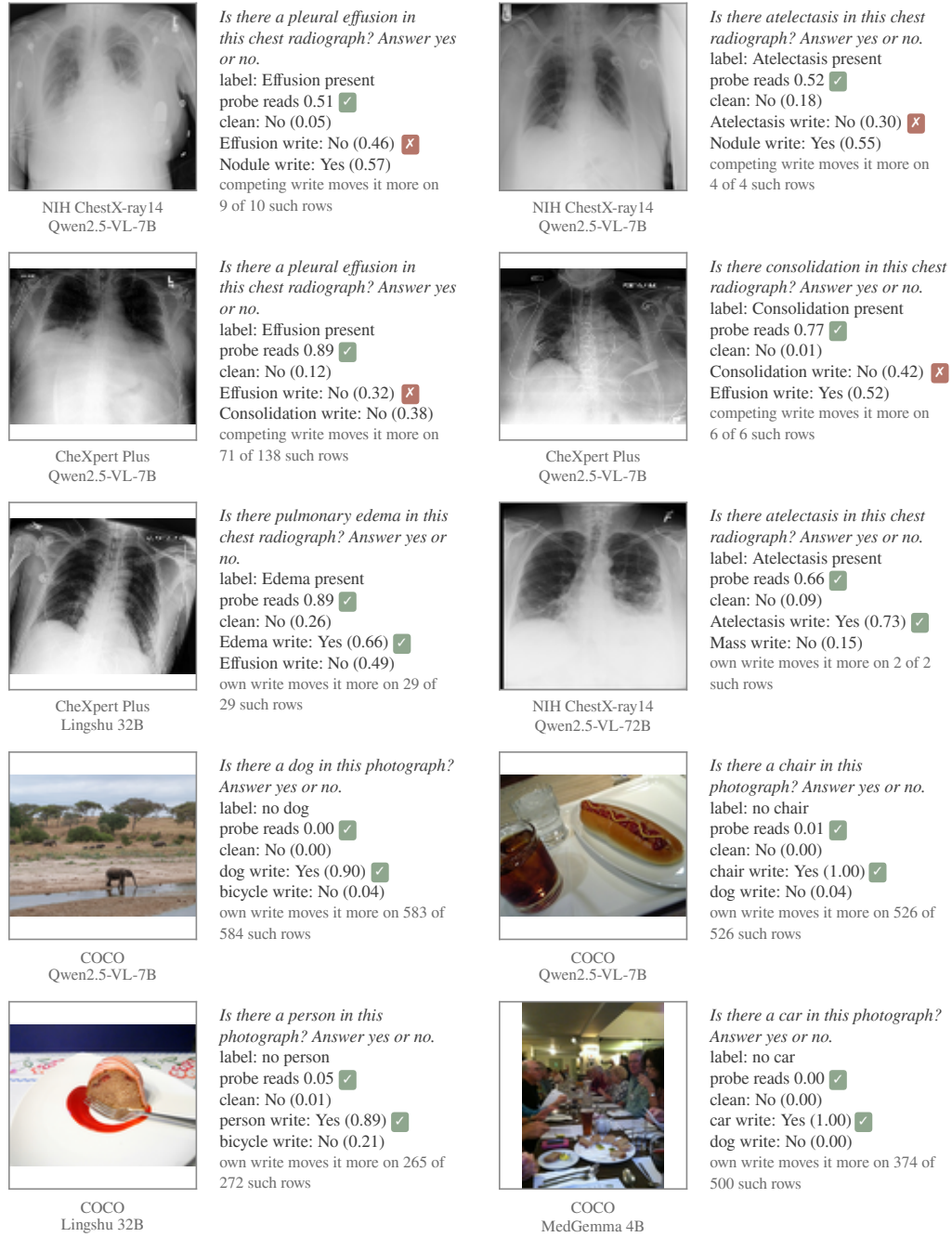


Figure 9: **Reading, answering, and writing on single images.** Per image: the question, the report label, the probe’s reading, the clean answer, and the answers under the own write and the strongest competing write at $\alpha = +0.25$. Beside the probe’s reading, a check or cross marks whether it agrees with the label; beside the own write, whether that write moves the answer more than the competitor. The grey line counts, among the block’s test rows the probe reads correctly and the model answers no when clean (on the chest sets, rows labelled present), those with the same ordering.

D Questions, Scores, and Notation

Templates and score. Each concept is asked in six templates that cross two wordings, *is there* (I) and *does the image show* (W), with three answer interfaces, yes/no (Y), A-present (A), and B-present (B). A template is named by its wording and its interface, and the primary template is IY. At the first answer position, ℓ_{yes} and ℓ_{no} are the maximum log probabilities among the valid single-token case and leading-space variants of each answer, and the registered score is $P(\text{yes}) = \sigma(\ell_{\text{yes}} - \ell_{\text{no}})$, where σ is the logistic function. Taking the best valid variant for each answer makes the score insensitive to which capitalisation or leading-space form receives the highest probability. The score measures the relative support for yes versus no at the first answer position, not the accuracy of a generated report. Questions are the rows of W and directions its columns, so changing a direction never changes the question being scored.

Notation. Table 17 collects the notation of the paper.

Table 17: Notation used in the paper.

Symbol	Definition
c, q, d	A concept; the yes/no question about a concept; a write direction. Questions and directions are indexed by the concept they belong to.
cell, block	A checkpoint–concept pair on one dataset; a checkpoint–dataset pair.
$\mathbf{h}_{it}, \mathbf{h}_i$	Output of the consumed block for image i and consumed token t ; its mean over the consumed tokens.
D	Hidden width of the consumed block.
R, s, β_c	Fixed $D \times 512$ Gaussian projection; training-row feature scales; logistic coefficients of concept c ’s probe.
$\hat{\mathbf{w}}_c$	Label direction: the probe’s logit gradient $R \text{diag}(s)^{-1} \beta_c$ at unit length (Equation 2).
a_q	Answer direction: the ridge fit of the clean answer margin $\ell_{\text{yes}} - \ell_{\text{no}}$, lifted in the same way.
α	Signed dose; $ \alpha $ is the write’s norm as a fraction of each token’s norm.
S	Probe selectivity: concept AUROC minus the mean AUROC of 20 random-label control probes (Equation 3).
$W_{q,d}$	Mean change in question q ’s $P(\text{yes})$ when direction d is written.
O_q	Ownership, $W_{q,q} - \max_{d \neq q} W_{q,d}$ (Equation 4).
O_q^m	Ownership computed on changes in the logit margin $\ell_{\text{yes}} - \ell_{\text{no}}$.
κ_d	Column selectivity, $W_{d,d} / \sum_q W_{q,d} $.
$F_{q,d}$	Mean over checkpoints of $\max(W_{q,d}, 0)$, the band width in Figure 3.
$\cos_\Sigma$	Whitened cosine under the training-feature covariance Σ (Appendix A.10).

E Joint bootstrap and simultaneous lower bounds

For a family of m statistics, let $\hat{C}_j$ be statistic j ’s estimate, C_{bj}^* its value in joint bootstrap draw $b \in \{1, \dots, B\}$, and τ_j its bootstrap standard deviation with denominator $B - 1$. The centred studentised draws, the common critical value, and the simultaneous lower bounds are

$$Z_{bj} = \frac{C_{bj}^* - \hat{C}_j}{\tau_j}, \quad c_{.95} = \mathcal{Q}_{.95} \left(\left\{ \max_{1 \leq j \leq m} Z_{bj} \right\}_{b=1}^B \right), \quad L_j = \hat{C}_j - c_{.95} \tau_j, \quad (16)$$

where $\mathcal{Q}_{.95}$ is the empirical 95th percentile with linear interpolation. The draws are centred on the estimate, and each τ_j stays fixed across draws. A component with $\tau_j \leq 10^{-12}$ gets $Z_{bj} = 0$ but keeps its own τ_j in L_j ; every draw is retained. Upper bounds use the same $c_{.95}$ with every contrast’s sign reversed.

Verdicts. For the ownership family of 30 contrasts $W_{q,q} - W_{q,d}$, question q ’s simultaneous lower bound is the minimum of its five contrast bounds. The question has the advantage when that minimum is positive, a stronger competitor when any of its upper bounds is negative, and is unresolved otherwise. O_q itself receives a percentile interval instead: every draw recomputes the largest competitor, the two-sided interval takes the 2.5th and 97.5th percentiles, and the pilot’s Effusion supplement takes the 5th as its one-sided bound. The logit-scale grade (Appendix A.7) applies both rules to margin changes.

Draws. The campaign resamples the 600 test rows of each block in 2,000 shared draws (seed 2026090601). The pilot’s ownership matrix resamples 400 patients in 5,000 draws (seed 20260904), with the same patient indices for every question and intervention, and bounds its shared-alias statistics as a separate family of 12. The Mass confirmation resamples 200 new patients (seed 20260910), with a family of 20 Mass-minus-alternative contrasts across four coded prompts.

F Reproducibility specification

This appendix specifies each family’s consumed block, the consumed-token mask, the projection and probe, the control constructions, the cohort roles, and the protocol record at the registered protocol’s level of detail. The released code implements each component.

F.1 Consumed loci per family

The consumed block is the last visual block that feeds the consumer. The connector locus is the consumer’s output that enters the language model. We identify the block from each checkpoint’s own configuration when loading the checkpoint. Each block’s preflight record stores the identified block, its hidden dimension, its consumer, and any branch that bypasses it.

- Qwen2.5-VL, Lingshu, and Qwen3-VL: the merger reads the last block of the vision transformer, which has 32 blocks in Qwen2.5-VL and Lingshu, 24 in Qwen3-VL 4B, and 27 in Qwen3-VL 8B and 32B. A Qwen2.5-VL or Lingshu image contributes 576 patch tokens, which the merger joins in 2×2 groups into 144; a Qwen3-VL image contributes at most 1,024. The Qwen3-VL DeepStack mergers, which read earlier blocks, are not written.
- Gemma 3 and MedGemma: the projector reads the last of the 27 SigLIP encoder layers through the tower’s final layer normalisation, and consumes all 4,096 patch tokens of the 896×896 image.
- InternVL3.5: the projector reads the last InternViT encoder layer, of 24 at 8B and 14B and of 45 at 38B; the tower’s final normalisation is an identity in the released configuration. An image contributes 1,024 patch tokens from a single 448×448 tile.
- LLaVA-1.5 and LLaVA-Med: the projector reads the penultimate of the 24 CLIP encoder layers, and the final layer is computed and discarded. An image contributes 576 patch tokens.
- Llama 3.2 Vision: the projector input concatenates the last of the eight global-encoder layers with five intermediate layers of the local encoder, which bypass the consumed block. An image contributes 1,601 tokens, the CLS token and 1,600 patches, from its one real 560×560 tile.

Table 2 lists the consumed-token count of every checkpoint.

F.2 The consumed-token mask

The write in Equation 1 changes only the consumed tokens at the positions the consumer reads at the locus. We derive these positions for each batch from that batch’s processor outputs:

- Qwen2.5-VL, Lingshu, and Qwen3-VL: the tokens of all images in a batch form one sequence in which every image occupies one contiguous span, whose bounds follow from the image’s patch grid. At the connector locus the span is a quarter as long, one merged token per 2×2 patch group.

- Gemma 3 and MedGemma: all 4,096 patch tokens; the tower produces no CLS token. The number of written tokens must equal the number of image placeholder tokens in the language-model input.
- InternVL3.5: the patch tokens of the single 448×448 tile. The CLS token, which the consumer drops, is excluded. The number of written tokens must equal the number of image placeholder tokens.
- LLaVA-1.5 and LLaVA-Med: the 576 patch tokens. The CLS token, which the checkpoint’s feature selection drops, is excluded.
- Llama 3.2 Vision: the 1,601 tokens, CLS and 1,600 patches, of the one real 560×560 tile. The seven padding tokens of each tile and the three padding tiles are excluded, in agreement with the model’s own aspect-ratio and cross-attention masks.

Scoring stops with an error and no fallback when an image span does not exactly cover the locus, a mask length differs from the token count at the locus, an image has no consumed token, or one forward pass reaches the locus more than once.

Preflight check (C) verifies the mask. It captures the locus tensor on a clean pass and a pass steered at $\alpha = +0.25$ along the first concept direction. It reports the maximum absolute difference outside the mask and the maximum absolute changes in the consumer’s output and answer logits. Across all 75 blocks, the change outside the consumed tokens is 0 at both loci. The consumer’s output and answer logits both change. The same records show a bitwise-identical repeated clean pass (check A) and a bitwise-identical $\alpha = 0$ pass (check B) for every block.

F.3 Projection, lift, scaler, and probe

We construct the projection $R \in \mathbb{R}^{D \times 512}$ by drawing standard normal variates once from a PCG64 generator seeded with 0, dividing by $\sqrt{512}$, and storing the result in float32. Its entries are independent $\mathcal{N}(0, 1/512)$ variates. The seed is 0 for every checkpoint, dataset, and locus. D is the family’s hidden dimension. We neither normalise nor orthogonalise the columns of R . The Gram matrix $R^T R / D$ has full rank 512 in all 57 blocks that carry the displacement module, with eigenvalues 1.7×10^{-4} – 5.7×10^{-3} and condition number 5.3–33.2.

Projecting the token-mean read vector $\mathbf{h}_i$ gives features $R^T \mathbf{h}_i$. A featurewise standardiser fitted only on training rows supplies the mean $\boldsymbol{\mu}$ and scale $\mathbf{s}$. Every row is standardised as $(R^T \mathbf{h} - \boldsymbol{\mu}) \oslash \max(\mathbf{s}, 10^{-8})$, where $\oslash$ and $\max$ act elementwise. Each concept probe is an ℓ_2 -regularised logistic regression with inverse regularisation strength $C = 1$. We fit each probe by L-BFGS for at most 2,000 iterations with unweighted classes and seed 0, using training rows with a known label for that concept. For each of the two refit seeds, we resample training units with replacement, retaining all rows of each sampled unit at its multiplicity. We then refit the scaler and six probes using the same projection. The random family, shams, and control probes remain at seed 0.

F.4 Control constructions

Random directions. We draw one $119 \times D$ standard-normal matrix using a PCG64 generator seeded with 0 and normalise each row to unit ℓ_2 norm. Every question in the block uses the same 119 directions.

Coordinate-permutation sham. Immediately after the random draw, the same generator produces one permutation of $\{0, \dots, D-1\}$ per concept, in protocol concept order. We form concept c ’s sham by rearranging the coordinates of $\hat{\mathbf{w}}_c$ with concept c ’s own permutation. This preserves the direction’s coordinate multiset and unit norm. We write each question with its own sham.

Matched random-label probes. Each row has a type: view $\times$ sex $\times$ age decade on the chest sets, and aspect-ratio $\times$ area buckets on COCO. For control seed $k \in \{0, \dots, 19\}$, we traverse the types in sorted order and use a PCG64 generator seeded with k to draw one label per type uniformly from $\{0, 1\}$. If all types receive the same label, we flip the first. Every row of a type receives that type’s

label. We give each control probe the concept probe’s capacity and nuisance structure by fitting it on the same training rows and known-label subset with the same hyperparameters. A block fits control probes when its training rows span at least eight types. Selectivity is Equation 3. Its bootstrap over calibration units retains a draw when the real AUROC is finite and at least ten of the twenty control AUROCs are finite. We grade a cell for readability when at least 1,900 of the 2,000 draws are retained.

F.5 Cohort roles

Every chest image is a frontal study. One role holds more than one row per unit: the NIH training role, which takes every study of each training patient, 18,212 studies from 5,783 patients, 2,655 of whom contribute more than one and one of whom contributes 125. Every other role of every dataset holds one row per unit, and no unit appears in two roles of a dataset. We use the calibration role only to determine readability and answerability. We neither fit probes nor score writes on it. Every write uses the test role.

F.6 Protocol amendments

Every analysis the paper reports beyond the seven planned modules was added to the protocol file as a dated amendment after the first wave, and each amendment precedes the first result of the module it registers. Each amendment entry names its own rows, references, and decision rule before it names a result.

The fine-grained category selection rule. The rule that picks the six fine-grained COCO categories is stated in the protocol file in full, and it reads on the labels alone. A category is eligible when its training positives are at least as many as the weakest of the six existing concepts, when it has at least ten positives among the 400 calibration rows and at least ten among the 600 test rows, and when its mean relative instance area is below the median of the six existing concepts. Eligible categories are ranked by calibration positives in descending order, ties by mean relative instance area in ascending order, and remaining ties by category identifier; the first six are taken. The rule uses the label counts and the instance areas of the fixed cohorts and nothing else: no probe, no answer, and no write enters it. The protocol file carries the rule under the date of its amendment, and the module’s first result is later than that date, so the six categories and the two matching windows were fixed before any of them was scored.